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When OXXO Moves Into the Block: Evaluating the Impact of OXXO on Crime Activity in Bogotá

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Abstract

Although Bogotá, the capital city of Colombia, has experienced a significant reduction in crime since the 1990s, delinquency and insecurity remain major public concerns. OXXO, a conveniece store chain that has steadily expanded across the capital, may be an urban factor that is actively shaping Bogotá's criminal environment. This study employed TWFE and Callaway and Sant'Anna approaches with their respective event studies to estimate the effect of OXXO openings on local crime activity in Bogotá's Zonal Planning Units, and whether effects vary by crime type and time since opening. The most robust results from TWFE were homicide and sexual violence. They showed that UPZs with OXXO presence present a 5.9% increase in homicides, and a 16% reduction in sexual violence compared to UPZs without OXXO stores. Additionally, I found that both in the medium- and long-run OXXO stores do not appear to increase or decrease their impact on crime activity. These findings improve the understanding of how commercial establishments influence patterns of criminal activity in Bogotá, and suggest that policy resposes should exploit the positive externalities generated by OXXO stores, while mitigating the possible mechanisms that increase the risk of personal theft and homicide.

Keywords: Convenience stores, crime activity, Bogotá, OXXO, urban crime.
JEL codes: C21, C23, K42, R11, R14, R58.

*The code and data repository for this thesis can be accessed at https://github.com/sophiearisti/thesis_economics.git

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2 Introduction

Although Bogotá, the capital city of Colombia, has experienced a significant reduction in crime since the 1990s (J. F. Godoy et al., 2018; Rodríguez, 2017), delinquency and insecurity remain major public concerns. The implementation of preventive programs and policies by Bogotá's administrations since the 2000s, along with other measures targeting crime hotspots (J. F. Godoy et al., 2018), has not effectively deterred the high levels of personal theft and the growing perception of insecurity that have affected the daily lives of citizens in recent years (Cámara de Comercio de Bogotá, 2026b). This continued prevalence suggests that additional urban dynamics may shape patterns of criminal activity in Bogotá, and that alternative approaches and other possible factors to understanding crime in the city may still need to be explored.

One factor discussed in urban crime literature is the role of commercial establishments in shaping local patterns of criminal activity (Askey et al., 2018; Davis et al., 2021; Dugato et al., 2025; Nazzari et al., 2026; Wellford et al., 1997). Within this category of establishments, OXXO, a convenience store chain that has steadily expanded across Bogotá since 2009, can be a case study for exploring this phenomenon in the city (M. C. P. Godoy, 2025). In particular, due to their long operating hours, strategic locations in high-traffic areas and commercial corridors (Marcos, 2022), availability of basic goods, and constant customer flows throughout the day and night, the arrival of this chain may alter surrounding patterns of mobility and social interaction in ways that could influence nearby criminal activity (Dugato et al., 2025; Marcos, 2022).

There is a duality in the crime literature regarding the relationship between convenience stores and crime. On the one hand, these establishments have been identified as *crime generators* because they increase foot traffic and create incidental criminal opportunities (Dugato et al., 2025). Similarly, Wellford et al. (1997) classify them as robbery-prone commercial establishments due to the availability of cash on hand. On the other hand, Davis et al. (2021) offer a different viewpoint, arguing that convenience stores may also function as crime deterrents by increasing natural surveillance and generating *eyes on the street* through extended operating hours and improved lighting. These two lines of reasoning highlight the importance of analysing the role of convenience stores within the surrounding criminal environment, particularly because, as Dugato et al. (2025) argue, addressing the urban conditions that facilitate criminal behavior may be more effective for long-term crime reduction than traditional repressive approaches.

OXXO's footprint in Colombia is centralized, with over 300 of the more than 500 stores in the country concentrated in the capital city (M. C. P. Godoy, 2025). This high density within a single metropolitan area provides an "urban laboratory" to conduct a granular, intraurban crime analysis. By exploiting the staggered arrival of these establishments across Bogotá's Zonal Planning Units (UPZs), this thesis evaluates the causal impact of OXXO openings on local criminal activity. Specifically, I examine how these effects vary by crime category, and whether the impact intensifies or dissipates over time following the store's opening.

To estimate this impact, I constructed a quarterly panel dataset covering the period from 2015 to 2022, aggregated at the UPZ-level. The panel combines Colombia's public georeferenced crime records from the National Police's Statistical, Criminal, Contraventional, and Operational Information System (SIEDCO) with a georeferenced database of OXXO establishments that accounts for both opening and, when applicable, closure dates. Using this information, I calculated quarterly crime rates for each UPZ and identified the number of OXXO stores operating within each unit over time. An exception is personal theft and sexual violence, whose analyses are restricted to the 2015–2018 and 2015–2021 periods, respectively, due to data availability constraints. Consequently, the results for these crime categories, particularly personal theft, should be interpreted with caution because they rely on a shorter observation period than the remaining crime outcomes.

Additionally, I employed Two-Way Fixed Effects (TWFE) and the Callaway and Sant'Anna (C&S) estimator, along with their corresponding event-study specifications, as the main empirical strategies of this investigation. The TWFE model was useful for estimating the average relationship between the expansion of OXXO stores and crime rates across UPZs over time. However, because staggered treatment adoption may generate bias under treatment-effect heterogeneity, the C&S estimator was also implemented to obtain more robust estimates and assess the consistency of the TWFE results. Finally, event-study analyses were conducted to examine the dynamic evolution of treatment effects over time and to evaluate the parallel trends assumption for both empirical approaches.

At the end I find that the staggered entry of OXXO stores has a statistically significant effect on criminal activity across Bogotá. Through the TWFE model, I demonstrate that homicide and sexual violence are the only crime categories that exhibit a robust positive effect across specifications. And the TWFE estimates, after applying the appropriate robustness checks, indicate that OXXO presence increases personal theft and homicide, while reducing sexual violence and vehicle and motorcycle theft. Additionally, for dynamic treatment effects, I found that OXXO stores do not appear to significantly increase or decrease their impact on crime rates in either the medium- or long-run. These statistically significant estimates may be explained by different mechanisms across crime categories. The increase in personal theft may result from the greater concentration of potential victims in a single location, whereas the rise in homicides may be driven by OXXO's 24-hour operating format, alcohol sales, and role in facilitating nocturnal gatherings. At the same time, increased pedestrian activity may enhance natural surveillance, thereby reducing sexual violence and vehicle and motorcycle theft.

Despite extensive research on urban crime in Bogotá, no study has systematically examined the causal impact of convenience store proliferation, particularly OXXO, on local crime patterns. Current research has focused on public transport infrastructure (Bacares & Augusto, 2013, 2014; Moreno, 2005), large-scale police interventions (Blattman et al., 2017, 2021), alcohol sales restrictions (de Mello et al., 2013), and socioeconomic determinants (Escobar, 2012). One recent study examined the relationship between *Tiendas D1* hard-discount stores and crime (Jaramillo López et al., 2021), and found a statisically significant effect of D1 openings on criminal activity; nonetheless, the effect of OXXO on crime is hypothesized to follow a different pattern, as it represents a distinct retail model characterized by 24-hour operations, and a higher availability of alcohol and *grab-and-go* items, which may create different criminal opportunities compared to hard-discount formats.

In the next section I briefly introduce OXXO's expansion across Bogotá, and the city's crime rates over the years. Then, in section 4 I present the related literature about the spatial nexus of crime and convenience stores. In sections 5 and 6 I describe the data sources used and the general descriptive statistics. After that I explain in section 7 the econometric models used, and in section 8 I show the results obtained.

In section 9 I include some robustness checks related with the specification of some variables. Finally, Section 10 discusses possible mechanisms that may influence the results obtained.

3 Background

3.1 The expansion of OXXO in Bogotá

Although OXXO has been operating in Colombia for almost 17 years, it only consolidated itself as one of the main convenience store chains in the country after the pandemic (El Economista, 2026). In fact, in Bogotá the number of establishments rose sharply from 82 in 2020 to 384 in 2025, this represents an increment of over 368% in only five years. Figure 1 depicts this accelerated growth in Bogotá, as it shows the number of OXXO locations in the capital by year since its arrival. This OXXO expansion can be due to some of its intrinsic characteristics; for example, 95% of its establishments function in a 24-hour format (OXXO Colombia, 2026a), providing a competitive advantage by being the primary option of consumers whenever they need to satisfy spontaneous consumption needs (López, 2026).

The proliferation of OXXO stores began in the northern part of the city before spreading throughout the eastern region. Figure 2 illustrates this evolution, showing that OXXO is now present in nearly 70% of the Bogotá's Zonal Planning Units (UPZ), these being units that group neighborhoods based on socioeconomic stratification, land use, housing characteristics, and the living conditions of their inhabitants (Alcaldía Mayor de Bogotá, 2021). Furthermore, it can be observed that OXXO presence has a primary concentration in the northeastern neighborhoods of Bogotá. Currently, the only areas lacking OXXO establishments are in the far southern reaches of the city. These places correspond to UPZs with lower levels of economic performance, socioeconomic stratification, and urban transport connectivity (Bocarejo & Tafur, 2013; Guzman & Bocarejo, 2017; PRISM, 2025), where residents continue to rely heavily on traditional neighborhood stores (Rico-Pencue et al., 2021).

This expansion over time positively correlates with the inherent characteristics of northeastern Bogotá, where high traffic and spatial accessibility provide an ideal environment for the convenience store model (Marcos, 2022). Figure 3 illustrates this

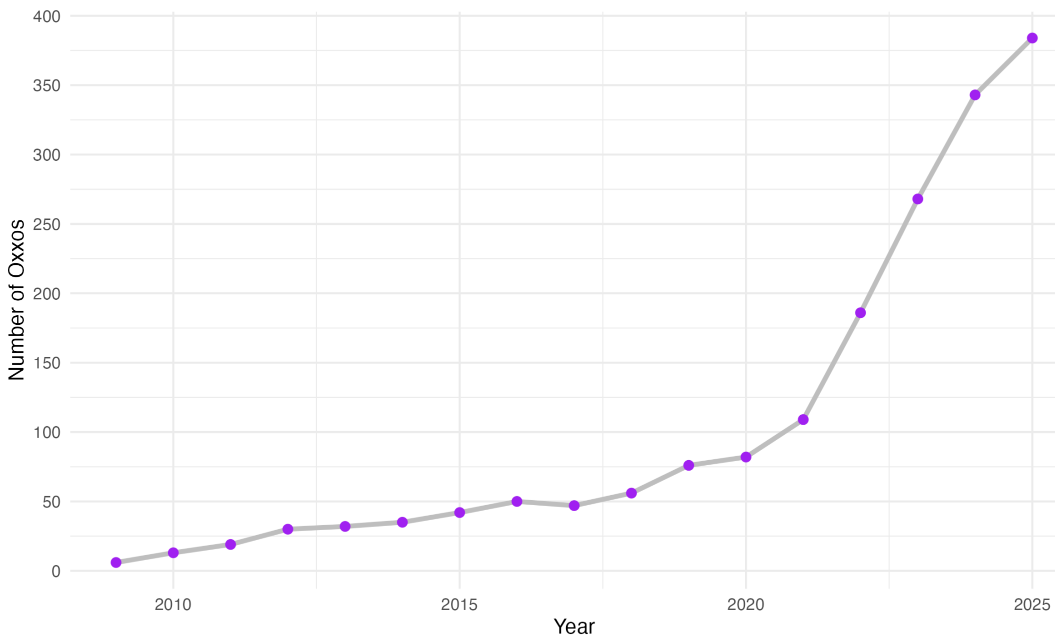


Figure 1: Yearly evolution of the number of OXXOs in Bogotá

behavior by mapping the city's arterial roads, main avenues, and Transmilenio stations (Bogotá's Bus Rapid Transit system); it demonstrates that as of 2025, OXXO locations tended to cluster along these major transport corridors. Section 6 provides a statistical analysis confirming the positive correlation between these variables and store locations.

OXXO locations are centrally administered by FEMSA with the purpose of standardizing processes, logistics, services, and products (OXXO Colombia, 2026a). This implies that store placement is strategically determined according to the company's broader commercial objectives rather than being randomly distributed across space. Such non-aleatory allocation may complicate econometric identification, since the characteristics that attract OXXO establishments may also affect crime behavior.

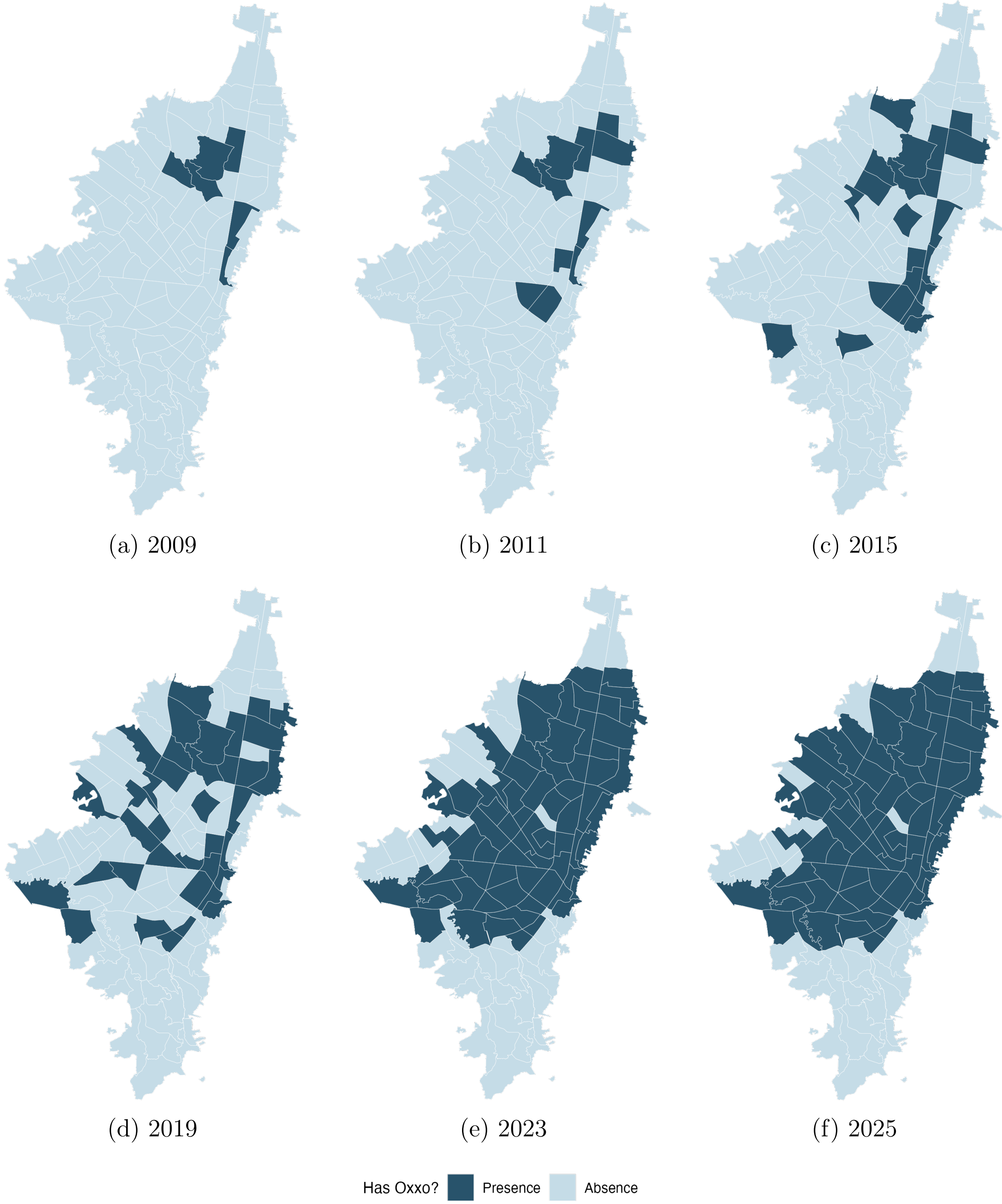


Figure 2: Evolution of the presence of OXXO by UPZ
Note: This map only indicates which UPZs have OXXOs, it does not consider the number of locations present in each Unit

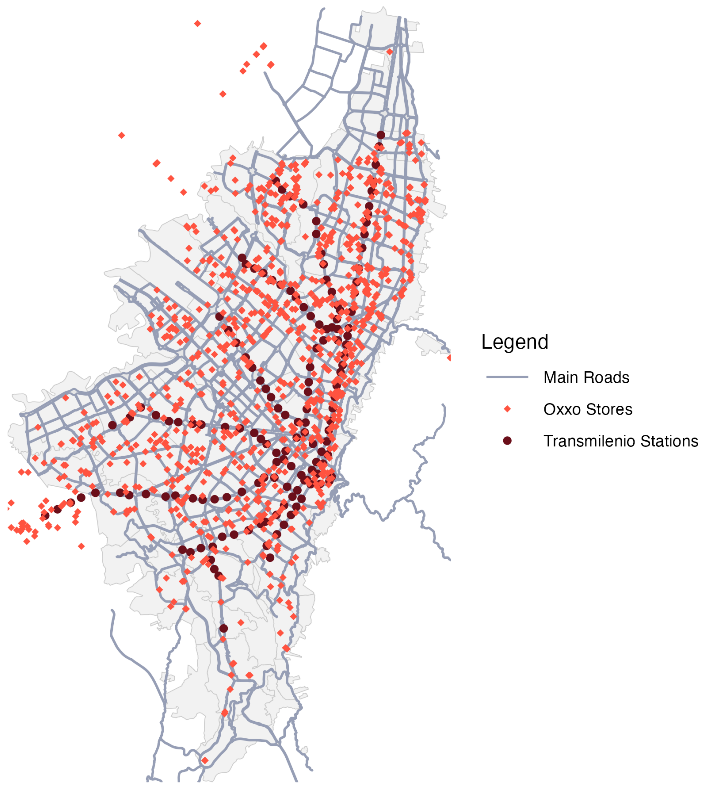


Figure 3: Bogota's OXXO stores, Transmilenio stations and main roads in 2025

The concern about endogenous OXXO placement, however, may be mitigated through the use of TWFE estimators, because many determinants of store location are either time-invariant or evolve only gradually over the sample period, allowing UPZ fixed effects to absorb a substantial portion of the selection process. Nonetheless, some bias may persist if an OXXO opening responds to time-varying local trends correlated with crime activity. For example, FEMSA may prioritize expansion into UPZs experiencing increases in commercial activity, population density, household income, pedestrian flows, or improvements in public security, all of which could independently influence criminal activity. In such cases, TWFE estimates can only partially capture the causal effect of OXXO on crime.

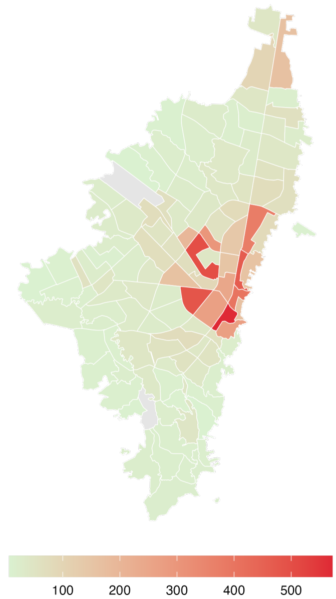
3.2 Criminality in Bogotá

During the 1980s and 1990s, Bogotá was among the most unsafe cities in Colombia, experiencing a surge in narcoterrorism and the assassination of political leaders; in fact, in 1991, it reached a rate of 1,287 crimes per 100,000 inhabitants, exceeding that of Medellín (1,160) and Cali (537) (Rodríguez, 2017). However, at the end of the 90s and beginning of the 2000s, crime started to decrease; for example, between 1993 and 2002 Bogotá saw an estimated 65% decrease in the homicide rate, while robbery fell by about 33%. This shift was driven by citizen culture programs, large-scale urban renewal projects, and institutional reforms implemented by the mayors of this period (Mockus, 2012; Sánchez et al., 2003). In recent years, however, urban crime has shown a fluctuating behavior. Following the 2019 covid pandemic, criminality had a rebound, with personal theft surpassing pre-pandemic levels in 2022 and 2023 (Policía Nacional de Colombia, 2026). By 2024, despite continued institutional efforts and a decrease in crime rates in the city (Policía Nacional de Colombia, 2026), 69.3% of households believed insecurity had raised (Cámara de Comercio de Bogotá, 2025).

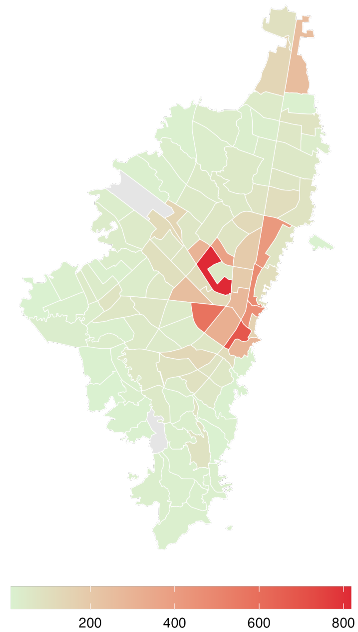
Currently, crime in Bogotá, as in most cities, tends to cluster in specific zones (Enriquez, 2014; Giménez-Santana et al., 2018). The most insecure Localities (a unit of territorial division that gathers many UPZ) are primarily located in the southeastern region of the city, a pattern associated with the presence of criminal organizations, high levels of informality, and deficiencies in human development (ProBogotá Región, 2024). However, this spatial concentration does not imply that all categories of crime peak in the same areas. While homicide rates generally follow this pattern (see Appendix D), personal theft exhibits a distinct geographic distribution, with higher concentrations in the northeastern sector of the city (ProBogotá Región, 2023).

Figure 4 illustrates this behaviour as it displays how personal theft rates distribute throughout Bogotá´s UPZs over the years (Appendix D shows how other crimes distribute). These maps also highlight that between 2015 and 2021 (and even later, as it can be noted in ProBogota's annual security reports) this tendency has not drastically changed. This persistence suggests that there are environmental characteristics that keep favoring robberies or attracting offenders. Among these characteristics are high pedestrian traffic, a dense concentration of commercial land use, and proximity to the city's main transit corridors.

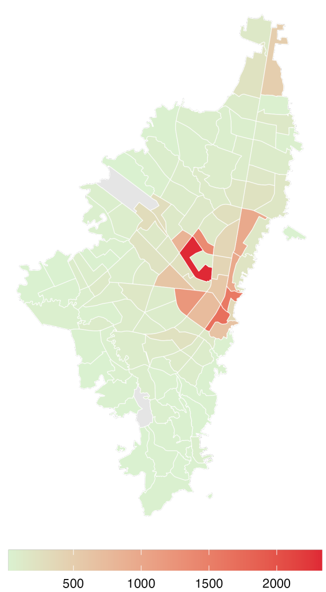
It is precisely within these high-activity environments that OXXO has focused its expansion. Consequently, for this research, factors that favor crime clustering must be taken into account. If OXXO stores tend to locate in inherently high-crime areas, it



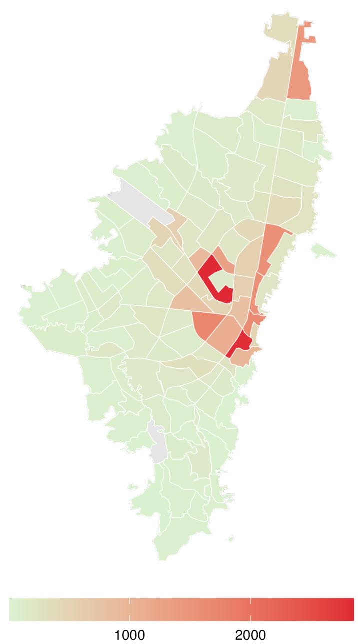
(a) 2015



(b) 2016



(c) 2017



(d) 2018

Figure 4: Evolution of theft rates by UPZ

Note: The theft rate per 10,000 inhabitants for each unit was calculated using rate smoothing techniques (Empirical Bayes smoothing). This approach aimed to reduce the influence of outlier UPZs that had remarkably high rates due to their small populations rather than high theft activity. Still, after this transformation, I had to exclude two UPZs, as they continued to show extreme values that overshadowed the crime activity of other areas. In fact, one UPZ is an airport, and the other is mainly rural and contains many storage facilities instead of dwellings.

becomes essential to assess whether their presence contributes to an increase in crime or whether both the stores and offenders are simply drawn to the same high-activity nodes. In this sense, OXXO locations may function either as crime generators or as crime attractors.

4 Related Literature

In the context of urban economics, spatial econometrics and crime literature, various studies have explored how built environment features and infrastructure shape crime activity in cities. A number of empirical analyses have investigated this by exploring the causal impact of urban public transit on crime. Xu et al. (2021), for example, assessed the impact of two new subway lines implemented in 2017 in ZG city, China, on thefts in the surrounding areas. By employing a Difference-in-Differences (DiD) framework, they found that in the short run, theft increased by 15%. Others, on the other hand, have explored the paper of new urban policies and infrastructure interventions on this topic. Domínguez and Asahi (2022) paper is one case, as they study the causal effect of ambient light on crime in Santiago, Chile, by using the *Daylight Saving Time* policy as a natural experiment. In this study they found that an increase of one hour of ambient light exposure between 7 and 9 reduced thefts by 30%. These studies emphasize that urban interventions do not just shift crime spatially, but actively reshape the rational choice of offenders by creating or eliminating environmental opportunities for illicit activity.

Retail chains can be considered as built environment features that may influence urban crime activity. While different studies have extensively explored their role, these previous inquiries have their own limitations. A substantial body of research dates back to the pre-2000s era, meaning that their findings do not contemplate the current urban dynamics of modern cities. Nonetheless, these foundational studies provide a critical perspective on the underlying mechanics of crime. For instance, early investigations focused on identifying the characteristics that influenced the propensity of convenience stores to be targeted for robbery. They identified that stores that operated for longer hours at night were more likely to be robbed, as there are fewer people around (D'Alessio & Stolzenberg, 1990). Others found that these types of stores were more prone to be robbed due to the lack of visible security such as police presence, security guards, or effective alarm systems (Dugato et al., 2025; Wellford et al., 1997). Although useful, they fail to account for convenience stores' role on crime of their immediate surroundings.

On the other hand, other recent publications have examined the influence of analogous commercial establishments on urban crime. While some of these studies do not focus exclusively on convenience stores, they utilize other retail outlets with similar operational profiles which can serve as proxies for what I aim to investigate. Askey et al. (2018) is an example, as the authors pooled data from fast-food restaurants and convenience stores in Seattle. They used sales revenue as a proxy for human activity to count the different levels of operational intensity across locations, and aimed to explain crime patterns in the immediate surroundings of these establishments. Their results indicate that stores with higher average sales were associated with with a 7.4% increase in expected crime counts. Conversely, Feng et al. (2019) contribute to this discussion by finding that alcohol outlets influenced the likelihood of aggravated assault and street robbery in New York City. Their findings reveal differential impacts depending on the type of outlet (if it is a bar, a grocery store, or other similar establishment); for instance, the authors highlighted that in Manhattan and the Bronx, proximity within two blocks of a grocery store increased the relative risk of street robbery by more than sixfold.

Another investigation is the one of Shin (2024). Shin (2024) used a DiD approach to investigate the impact of dollar stores on localized violent crime patterns in Chicago. To ensure the treatment and control groups were as homogeneous as possible, the author created localized areas using a 250-feet buffer for the treatment group and a 250-to-354-feet outer ring as the control. At the end, this research found that store openings lead to a 21.5% increase in violent crime (0.17 incidents per quarter), a result primarily driven by a 38% surge in robberies.

Although these studies offer a useful baseline of how convenience stores, and specifically OXXO, impact nearby crime, they present an incomplete perspective as they exclusively explored this nexus in cities of high-income nations. To build a more robust theory, these hypotheses must be tested within Latin American cities which have particular urban dynamics. Many papers argue that convenience stores are normally located in socio-economically disadvantaged areas where poverty, social disorganization, and lack of community resources exacerbate crime risks (Dugato et al., 2025; Shin, 2024), but in cities like Bogotá the situation is not analogous. OXXO in Bogotá, for instance, does not follow this rule; instead, it prioritizes opening locations in high-foot-traffic areas, mixed-use commercial corridors, and places with immediate spending capacity (Marcos, 2022), areas that are not often found in low-income neighborhoods (Cámara de Comercio de Bogotá, 2026a).

In Latin America, the economic literature regarding convenience stores remains limited. Recent investigations have explored their causal role on various labor market indicators (Delgado et al., 2024; Marcos, 2022). However, despite this growing interest in labor dynamics, the spatial relationship between these establishments and urban security in the region has been largely overlooked. Specifically in Bogotá crime literature has primarily focused on public transport, general socioeconomic urban caracteristics, and policy intervention impact on crime (Bacares & Augusto, 2014; Moreno García, 2005).

The investigations that have actually explored this topic in Bogotá are counted. For example, Jaramillo López et al. (2021) used the same method as Shin (2024) to study the effect of D1 hard discount stores on the crime around the areas where they open in Bogotá. Although D1s are not convenience stores per se, they attract a similar type of customers and and often share similar spatial locations with OXXO outlets. By analyzing the period of 2016 to 2019 and employing the same information utilized in this thesis (within SIEDCO infromation system), the authors found that D1 openings caused a 33.4% average reduction in crime density per block. They attributed this impact to the 'eyes on the street' phenomenon, suggesting that the increased pedestrian flow acts as a form of informal guardianship that deters criminal activity (Jacobs, 1961; Jaramillo López et al., 2021).

As noted, the body of literature that explores the impact of convenience stores on crime in Latin America is reduced, and virtually non-existent in the Colombian context. In this thesis I aim to fill this gap by evaluating the impact of OXXO store openings on local crime in Bogotá. Given their rapid expansion and market dominance, OXXO stores serve as a primary actor and a representative proxy for modern convenience retail in the city. Additionally, this analisis is particularly relevant in a context where crime remains a public policy priority and a growing concern for Bogotá's inhabitants.

5 Data used and unit of analysis

I use UPZs as the unit of analysis because they constitute a feasible and consistent territorial division of Bogotá. The city maintains continuity in the availability of data aggregated at the UPZ level across time, and security policies and interventions are frequently designed and implemented using this geographic scale (García, 2016). Additionally, UPZs preserve a certain degree of urbanistic and socioeconomic homogeneity by grouping neighborhoods with similar characteristics, while also providing an intermediate level of aggregation between localities and street blocks (*manzanas*).

For the results of this study I employ four main data sources (see Appendix A). For measuring crime rate in Bogotá I used the georeferenced crime events stored in SIEDCO [1] spanning from 2015 to 2022. SIEDCO is Colombia's official Police database that consolidate real-time citizen reports and administrative records of criminal activity. Here, for each incident recorded the system saves details including type of crime (e.g., homicide, personal theft, commercial burglary, or cell phone theft), the precise geographic coordinates of the event, date and time of the incident, and the victim's gender. With this information, I aggregated the crime data at the UPZ-quarter level, categorizing incidents by offense type. To ensure comparability across quarters and UPZs, I normalized these counts using annual population estimates for each UPZ. This allowed for the construction of a crime rate per 10,000 inhabitants, as shown in this equation:

$$crime\_type\_index_{upz,quarter} = \frac{total\_incidents_{upz,quarter}}{population_{upz,year}} * 10,000; \quad \forall upz, quarter, year \tag{1}$$

I smoothed the rates using an Empirical Bayes technique to reduce the influence of extreme values in UPZs with small populations. Nonetheless, Appendix B shows how results would look without this transformation. There are some differences after applying the changes, but they are not large enough to reverse the direction of the effects, and remain broadly consistent with those presented in the Results section, with the exception of homicide, whose statistical significance is sensitive to this choice.

[1] Access at https://oaiee.scj.gov.co/agc/rest/services/Tematicos_Pub/SIEDCO_Delitos_Pub/FeatureServer

The period from 2015 to 2022 is appropriate for this analysis, as it encompasses both the year of its initial steady expansion and the first years of its explosive growth in Bogotá. This timeframe provides sufficient temporal and spatial variation, as a larger number of UPZs enter treatment during the study period rather than before it. Before 2015, the lack of new store openings in other UPZs would result in an insufficient number of newly treated units, limiting the identifying variation necessary for using TWFE.

Personal theft and sexual violence are the only crime categories analyzed over shorter periods, covering 2015–2018 and 2015–2021, respectively, because the National Police has not made data for these offenses publicly available beyond those years. Consequently, the results for these crime categories, particularly personal theft, should be interpreted with caution, as they are based on a shorter observation period than the other crime categories.

Additionally, I used data from the Unified Business and Social Registry (RUES)[2], Colombia's central business registry, to identify the opening and closing years of each OXXO store. I then used the Google Maps API to obtain the precise georeferenced location of each establishment. Although concerns may arise regarding the accuracy of this matching process, particularly for stores that are no longer in operation, I manually verified each location using official OXXO sources. As a result, 95% of establishments were correctly assigned.

I followed the same procedure with D1, Justo&Bueno and ARA hard-discount chains. I considered these brands because they often share similar spatial locations with OXXO stores; in fact, retailers tend to cluster in certain areas to take advantage of agglomeration benefits (Konishi, 2005; Seong et al., 2022). Also, given that prior research suggests hard-discount stores influence both crime and commercial activity (Delgado et al., 2024; Jaramillo López et al., 2021), their inclusion as control variables is important to evaluate the validity of my results .

Finally, I use the 2007 Bogotá Quality of Life Survey[3] to characterize the baseline socioeconomic conditions of treated and never-treated UPZs before OXXO entered the

[2] Access at https://www.rues.org.co

[3] Access at https://www.dane.gov.co/index.php/estadisticas-por-tema/salud/calidad-de-vida-ecv/encuesta-nacional-de-calidad-de-vida-2007-bogota

Table 1: Distribution of UPZs and OXXO stores over time

Year	**Not yet treated**	**Treated**	**Total OXXOs**
2015	90	22	38
2016	89	24	46
2017	87	23	46
2018	87	28	55
2019	81	37	74
2020	72	40	81
2021	70	49	108
2022	51	59	173
Total UPZs	**110**	–	–

Note: This table presents the annual count of UPZs by treatment status and the total number of OXXO stores in Bogotá. A UPZ is treated if it has at least one store. In 2017, one UPZ stopped being treated, as the only OXXO store it had closed; this UPZ was excluded from the regressions, since store closures are not contemplated in either the TWFE or the C&S specifications. Nonetheless, the total number of OXXOs remained the same across 2016 and 2017, because another OXXO store opened in a different UPZ that same year.

market in 2009. These variables are available only at the Locality level because they are not publicly disaggregated at the UPZ level. I also use data from Mapas Bogotá to construct time-invariant socioeconomic and environmental variables at the UPZ level. Although I used these variables to describe the non-random spatial distribution of OXXO stores, they are absorbed by the two-way fixed effects (TWFE) specification and therefore are not separately identified in the regressions.

6 Descriptive Statistics

First, it is important to illustrate the preliminary differences between controlled and treated UPZs, not only in terms of crime activity, but also regarding socioeconomic and accessibility variables. Table 1 presents the staggered arrival of OXXO stores across UPZs per year between 2015 to 2022. This Table shows that there was an increase in treated UPZs starting in 2019, while a substantial number of control units remained available for comparison over the years.

This variation shown may not be as drastic as OXXO stores growth depicted in

Figure 1. This suggests that some UPZs host a disproportionately higher number of establishments in order to account for the overall increase in OXXO stores across Bogotá. Moreover, this raises the question of whether treatment intensity should be explicitly considered in the analysis. Although Figure 5 does not provide a detailed spatial distribution of OXXO stores at the UPZ level over time, it illustrates the evolution of the average number of stores per cohort. The Figure shows that, in early periods, differences between cohorts were relatively small (less than one store on average), whereas in later periods these differences became more pronounced. This pattern suggests that variation in treatment intensity may play a role in shaping the estimated relationship between OXXO presence and crime, and therefore warrants further evaluation.

On the other hand, Figure 6 presents preliminary evidence of the behavior of different types of crime rates across the periods. Specifically, they report the difference in mean crime rates between treated and not-yet-treated UPZs for each quarter, along with an indication of whether these differences are statistically significant at the 5% level. For instance, panel 6a shows that, for personal theft, there was a statistically significant difference between groups during all quarters. This indicates that, on average, treated UPZs exhibited higher crime rates than those not yet treated.

These results are consistent with expectations, as personal theft is the crime category most likely to be positively associated with the presence of OXXO stores. This finding aligns with the hypothesis that such establishments may function as crime attractors by increasing the flow of suitable targets. However, this pattern may also reflect the possibility that OXXO stores are disproportionately located in UPZs with inherently higher theft rates.

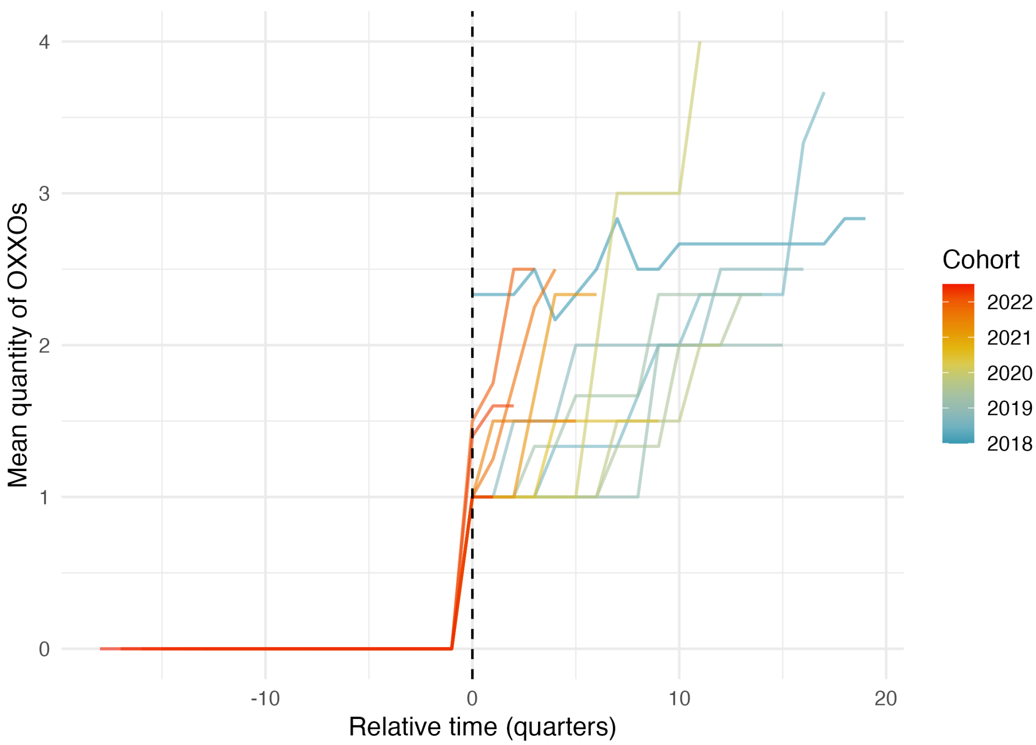
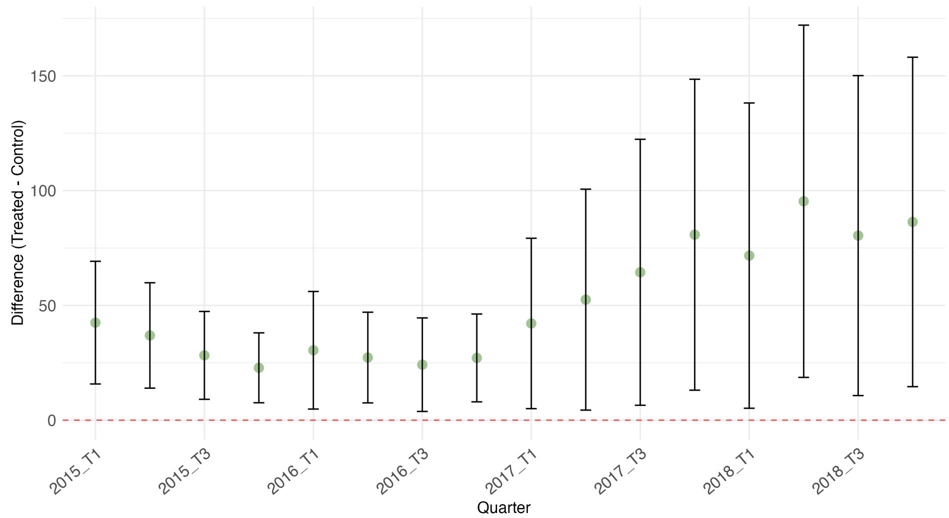


Figure 5: Treatment Intensity: Evolution of Average OXXO Quantity by Cohort

Note: The figure displays the average number of OXXO stores per UPZ, centered at the first quarter of entry ($t = 0$). Each line represents a different opening cohort (quarter-year). The color gradient indicates the chronological order of the cohorts. The dashed vertical line marks the start of the treatment.



(a) Personal theft rate

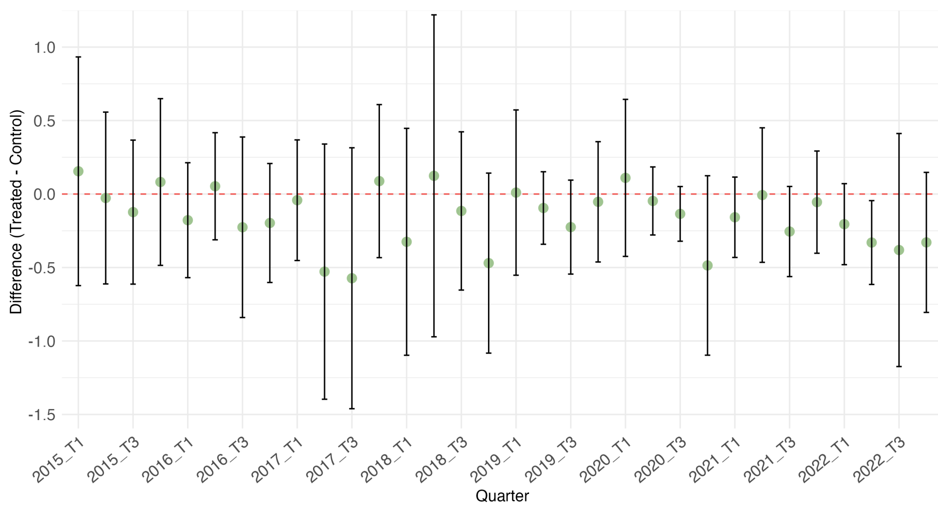
Figure 6: Difference in Means Analysis (Part I)

Moreover, the other graphs indicate that differences for other types of crime are not generally statistically significant. This fits the pattern expected, as crimes such as sexual violence (panel 6c) or vehicle and motorcycle thefts (panels 6d and 6e) are less likely to be influenced by OXXO presence. These offenses often rely on different spatial and social dynamics, such as secluded areas (Summers, 2014) or long-term parking patterns (Clarke & Mayhew, 1994), which may remain largely unaffected by convenience retail expansion.

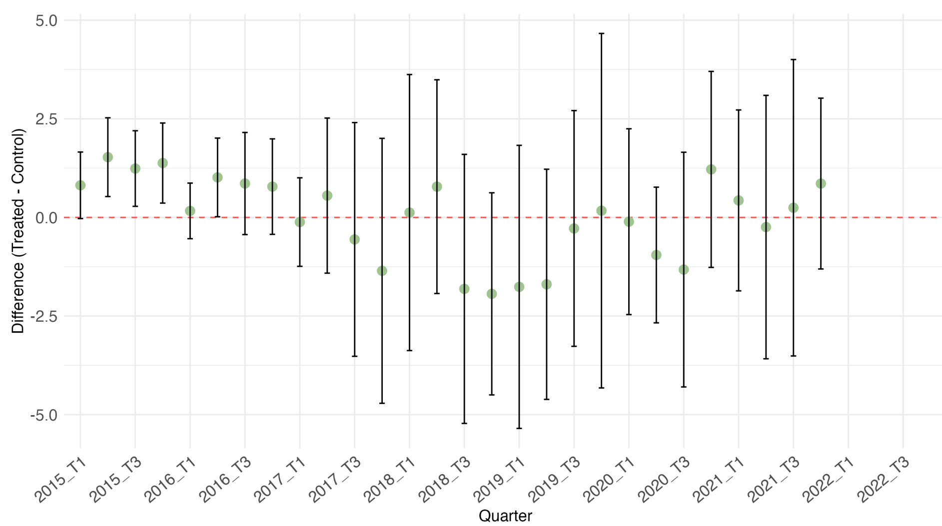
Table 2 also shows preliminary differences between treated and never treated groups. I chose these variables to help to visualize that OXXOs are located in areas where there is higher traffic, accessibility, quality of life, and purchasing power. Moreover, these results indicate that potential market size may not be adequately captured by population measures alone (such as total inhabitants or household size in 2007), but rather by indicators of economic activity and consumption capacity[4]. Since OXXO's product mix consists primarily of non-essential daily goods and convenience items rather than vital necessities, this retail chain prioritizes locations with *high-income transit* rather than

[4]Note that although the unit of analysis in this paper is the UPZ, the baseline covariates in Table 2 are measured at the Locality level, since disaggregated UPZ-level data for this type of information is not publicly available. Because each Locality contains multiple UPZs, this introduces some within-Locality measurement error: treated and control UPZs sharing the same Locality are assigned identical baseline values, even if their true underlying conditions differ. Despite this coarser level of aggregation, differences between treated and control units remain detectable even at the Locality level.

just high residential density. These results are consistent with what was confirmed by Marcos (2022), where convenience store chains in Mexico were found to be distributed by similar characteristics.



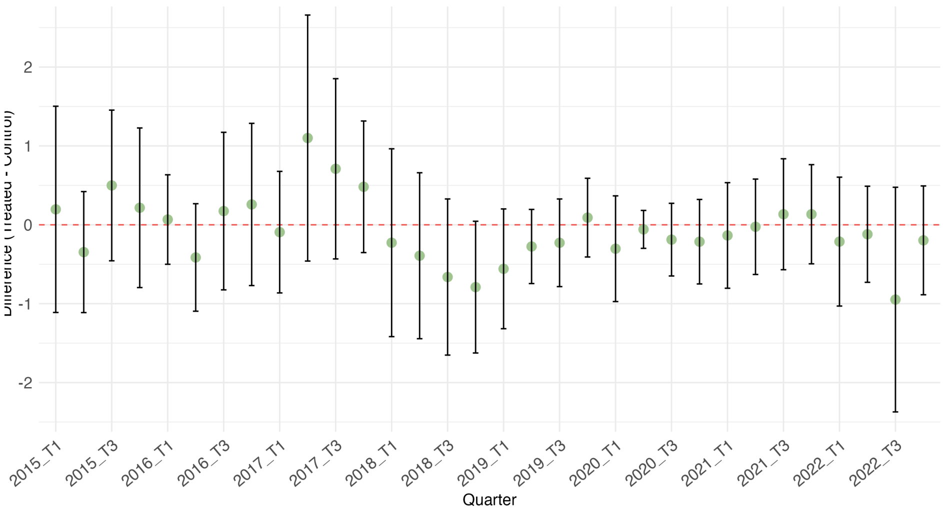
(b) Homicide rate



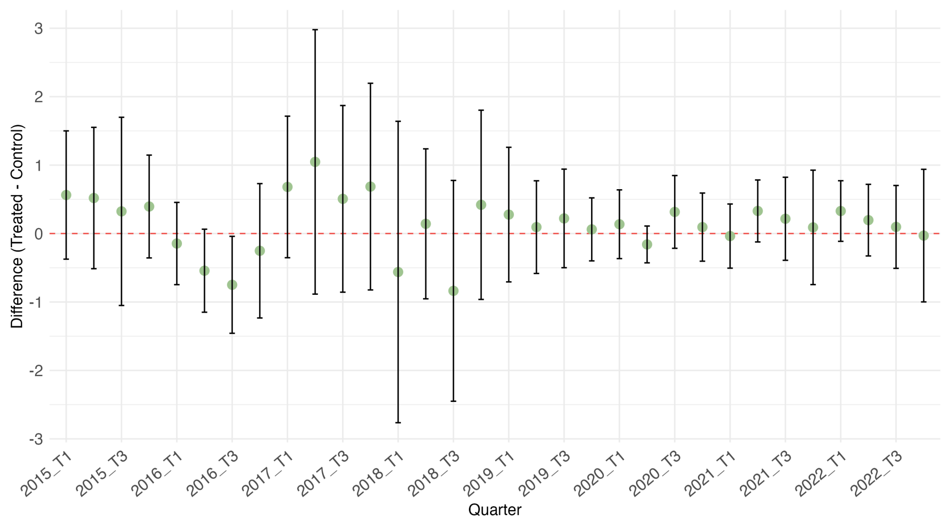
(c) Sexual Violence rate

Figure 6: Difference in Means Analysis (Part II)

Note: Sexual violence do not present differences in the last three quarters because SIEDCO's original data did not report cases of this type during this period.



(d) Motorcycle Theft rate



(e) Vehicle Theft rate

Figure 6: Difference in Means Analysis (Part III)

Note: This figure displays the difference-in-means tests for various crime categories across Bogota's UPZs. The confidence intervals represent 95% significance. If the intervals do not cross the zero line (red), the difference between treated and control units is statistically significant.

I also evaluated whether there was indeed a tendency for hard discount stores, such as D1, Ara, and Justo & Bueno, to locate near OXXO establishments. In Appendix C, I show some of the results, and in general find that for all years these hard discount stores were significantly more prone to locate in UPZs treated with OXXO than in those without. With this evidence, I included these chains as regressors because they not only influence OXXO's presence but also have a probable influence on the dependent variables, as shown by Jaramillo López et al. (2021) in their investigation. However, I included them in the regression as baseline covariates, meaning that I measured them only in the quarter preceding treatment to avoid potential confounding effects.

Table 2: Differences of means between treated and never treated for fixed and baseline controls

	Never treated	**Treated**	**Difference of means**
Baseline			
Inhabitants by Locality in 2007	515.290	486.391	−28.899
	(41.522)	(44.392)	
Household Size by Locality in 2007	3.681	3.331	−0.350***
	(0.051)	(0.048)	
Index of Quality of Life by Locality in 2007	87.400	91.529	4.129***
	(0.151)	(0.216)	
Log of Average Monthly Expenditure by Locality in 2007	13.652	14.086	0.434***
	(0.533)	(0.372)	
Fixed controls			
Average socioeconomic classification (estrato)	2.162	3.352	1.190***
	(0.135)	(0.119)	
Nearby transmilenio stations	2.667	6.034	3.367***
	(0.445)	(0.552)	
Access to arterial roads	29.549	43.831	14.281***
	(3.893)	(2.759)	
UPZs	51	59	110

Note: Significance: *** p<0.01, ** p<0.05, * p<0.1. Nearby TransMilenio stations and access to arterial roads are computed as the count of transport infrastructures located within the UPZ or whose 400-meter buffer zone intersects the UPZ boundaries.

Although I considered hard discount chains as factors that may bias the estimation, other unobserved variables could also affect the results. For instance, public works can reduce pedestrian and vehicular traffic for certain periods within the analysis; a new park may attract different demographic flows; and the presence of other commercial establishments, such as bars, nightclubs, or grocery stores, may impact the estimation of the causal effect, especially if they open or close during the period of study. Nonetheless, some of these variables may remain constant from 2015 to 2022; for example, the socioeconomic classification of a given block or UPZ typically changes very infrequently. Furthermore, other establishments may be controlled for by unit fixed effects, as they tend to be located in areas that maintain their core characteristics; for example, nightclubs often open in places already recognized as nightlife districts.

7 Empirical strategy

This research explores the causal effect of OXXO openings on local crime activity in Bogotá with two econometric models. The first one is TWFE, which takes into account that the the arrival of OXXO stores occurs in a staggered manner. Additionally, it inherently controls for fixed effects by unit (UPZ) and time (quarter). The former is useful for capturing time-invariant unobserved characteristics of each UPZ (such as geographic location, historical infrastructure or fixed socieconomic and urban characteristics during the period of analysis), while the latter absorbs common shocks that affect all units simultaneously (such as city-wide security policies or macroeconomic fluctuations). Although TWFE is well suited to estimating the Average Treatment Effect on the Treated (ATT) of OXXO store openings on crime activity, two identification assumptions are needed to prevent bias. These will be considered later, but let us first introduce the basic regression form of TWFE:

$$Y_{i,t} = \alpha_i + \alpha_t + \beta^{twfe} D_{i,t} + \gamma X_{i,t-1} + \varepsilon_{i,t} \tag{2}$$

Where $Y_{i,t}$ represents the crime rate for UPZ i in quarter t, this rate corresponds to a specific crime, either personal theft, homicide, sexual violence, theft to vehicle or motorcylce. $D_{i,t}$ is a binary indicator that equals one if UPZ i has at least one OXXO store in quarter t , and zero otherwise. Furthermore, as Figure 5 suggests that intensity of treatment may be a factor that influences crime. Section 8 also evaluated $D_{i,t}$ as the number of OXXO establishments that an UPZ has at a certain quarter. Also, $X_{i,t-1}$ represents the dummies for the hard-discount stores regressors (D1, Ara, and J&B), lagged by one quarter, to prevent confounding effects with OXXO's presence. Finally, α_i and α_t are the unit and time fixed effects respectively.

A critical identification assumption for the TWFE estimator is the homogeneity of treatment effects. This implies that these effects must be constant among treatment cohorts and over time (de Chaisemartin & D'Haultfœuille, 2023). This assumption may be violated if treatment effects vary based on UPZs' characteristics or if the impact of an OXXO opening on crime evolves as the store becomes established in the area. Consequently, to diagnose the potential bias arising from these heterogeneities, I employed the Goodman-Bacon decomposition (Goodman-Bacon, 2021). This theorem decomposes the β^{twfe} as a weighted average of all possible 2×2 difference-in-differences comparisons among cohorts, and the primary concern in this decomposition is to avoid having negative weights as they indicate that the estimator may be biased from heterogeneous treatment effects (Goodman-Bacon, 2021).

Parallel trends is the other condition to guarantee an unbiased ATT, as it ensures that, in the absence of treatment, the average outcomes of the treated and control groups would have followed the same trajectory over time. To evaluate the validity of this assumption, I implemented an event study design. This model requires no anticipation to treatment, and in this context it is highly unlikely that perpetrators or residents altered their behavior in expectation of an OXXO opening, as these commercial establishments are typically not publicized far enough in advance to influence local criminal dynamics beforehand. The event study is estimated by using the following equation:

$$Y_{i,t} = \alpha_i + \alpha_t + \sum_{k=-K}^{-2} \gamma_k^{lead} D_{i,t}^k + \sum_{k=0}^{L} \gamma_k^{lag} D_{i,t}^k + \varepsilon_{i,t} \tag{3}$$

Equation (3) allows analysis of the significance of the coefficients γ_k^{lead}, which evaluate evidence of parallel trends before the arrival of OXXO. Additionally, the coefficients γ_k^{lag} reflect the temporal dynamics of the treatment's impact.

Nonetheless, this model still needs to be compared with another that inherently solves the treatment heterogeneity bias. Thus, for robustness purposes, I estimated the Callaway & Sant'Anna (C&S) model. The objective of this comparison is to determine if C&S estimates and event study align with the TWFE results. A similar behavior between the two models would reinforce the validity of the TWFE estimates and suggest that the possible bias of heterogeneous effects is not a concern (Callaway & Sant'Anna, 2021; Rios-Avila et al., 2021).

The C&S approach accounts for treatment effect heterogeneity by estimating ATTs by cohort and period, subsequently combining these effects through a weighted average (Callaway & Sant'Anna, 2021; Rios-Avila et al., 2021). To achieve this, the model utilizes the following doubly robust estimator for the ATT of each cohort g at time t:

$$ATT(g,t) = \mathbb{E}\left[\left(\frac{G_g}{\mathbb{E}[G_g]} - \frac{\frac{p_{g,t}(X)(1-D_t)}{1-p_{g,t}(X)}}{\mathbb{E}\left[\frac{p_{g,t}(X)(1-D_t)}{1-p_{g,t}(X)}\right]}\right)(Y_t - Y_{g-1})\right] \tag{4}$$

Where t is a particular time period, G_g is a binary indicator for whether an observation belongs to cohort g, D_t is the treatment status at time t, and $(1 - D_t)$ identifies observations that have not yet been treated, serving as the control group. Additionally, the approach I chose uses Inverse Probability Weighting (IPW) to assign greater weight to control observations that more closely resemble the treated cohort G_g. In this case, the IPW accounts for all the controls included in the regression; this means it will only be relevant when evaluating the influence of hard-discount stores as a robustness test.

Subsequently, to aggregate all $ATT(g,t)$ estimates into a single summary parameter, the authors propose the following aggregation strategy:

$$\theta = \sum_{g \in \mathcal{G}} \sum_{t=2}^{\mathcal{T}} w(g,t) \cdot ATT(g,t) \tag{5}$$

Where $w(g,t)$ represents the weights determined by the researcher. In this analysis, I employ a simple averaging scheme where $w(g,t)$ is the same for all treated periods.

One additional consideration that warrants mention is that, due to the nature of both models, I decided to exclude from the estimation all UPZs that stopped being treated during the period of analysis, as well as those that were always treated. The first type of unit (which applied to only two UPZs) is not contemplated in the assumptions of either model, and its inclusion may affect the absence of heterogeneous treatment effects, as a change of state from treated to control will likely change crime activity as well. Moreover, the always-treated UPZs, as I will emphasize again in the next section, are inherently excluded from the C&S model, and for the TWFE they can bias the results due to the lack of a comparison group. Thus, to maintain the same universe of analysis for both models and prevent any bias, they were also excluded from the regressions.

8 Results

For this section I excluded the always-treated UPZs from the estimations because their inclusion biases the results, particularly for personal theft, where the TWFE specification for intensity of treatment yields substantially larger treatment effect estimates than expected and the event study pattern differs markedly from the C&S ES estimates. This bias arises because always treated UPZs do not have an appropriate comparison group and are systematically different from the rest of the sample. Their crime rates are persistently high and are more likely to attract OXXO stores due to their underlying characteristics, such as high levels of floating population, and their role as commercial hubs (Observatorio de Desarrollo Económico de Bogotá, 2022). As a result, the TWFE estimator is unable to disentangle these pre-existing differences from the effect of treatment. Appendix F reports the estimates obtained when always-treated UPZs are included in the sample, illustrating the resulting bias and facilitating comparison with the results presented in this section.

An additional aspect worth mentioning is that personal theft has to be analyzed with caution, not only due to its shorter period of analysis, but also because 65% of its original 26 treated UPZs are always-treated, leaving only nine UPZs that received treatment after 2015-Q1. This means there are few genuinely staggered treated units, which limits identification under staggered adoption, since there is little variation in treatment timing across quarters; this affects both TWFE identification and reliability.

Table 3 shows the estimates of equation 2 for each crime category. With the exception of homicide and personal theft, the estimated coefficients suggest that OXXO presence is associated with a reduction in crime. However, only the coefficients for homicide and sexual violence are statistically significant. The results indicate that the arrival of an OXXO store in a UPZ is associated with an increase of 0.084 homicides and a decrease of 0.44 cases of sexual violence per 10,000 inhabitants relative to UPZs without an OXXO store. Compared with the mean homicide and sexual violence rates in the never-treated group, these estimates correspond to a 16.97% increase and a 20.3% decrease, respectively. These findings are contrary to my prior expectations. I anticipated that sexual violence would not exhibit a statistically significant effect, whereas I expected personal theft to do so. And, if homicide was also statistically significant, I expected both crime categories to increase following the arrival of an OXXO store.

Table 3: Effect of OXXO Store Openings on Different Types of Crime Rates

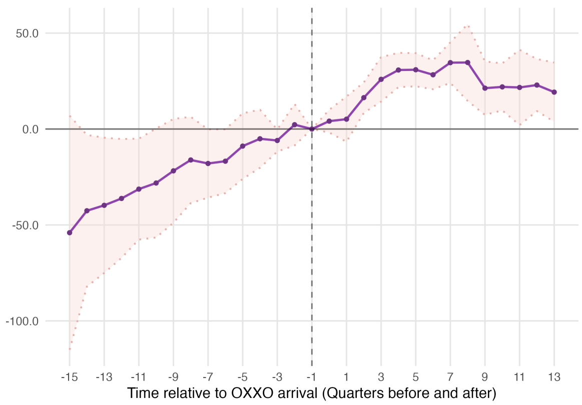
Crime rates	Personal Theft (1)	Sexual Violence (2)	Homicide (3)	Motorcycle Theft (4)	Vehicle Theft (5)
Has OXXO	2.331	-0.440^{**}	0.084^{**}	-0.085	-0.132
	(7.141)	(0.199)	(0.034)	(0.080)	(0.087)
Constant	30.96^{***}	2.199^{***}	0.459^{***}	1.523^{***}	1.370^{***}
	(0.263)	(0.023)	(0.005)	(0.012)	(0.013)
Observations	1,440	2,520	2,880	2,880	2,880
Treated (UPZ)	9	30	40	40	40
Never treated	81	60	50	50	50
R-squared	0.660	0.416	0.395	0.611	0.582
Mean never treated	55.531	2.167	0.495	1.643	1.153

Note: Standard errors in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.
TWFE considers fixed effects by unit and time, with standard errors clustered by UPZ. The dependent variable is the smoothed crime type rate per 10,000 inhabitants per UPZ. Moreover, the treatment variable is binary and indicates if the UPZ has at least one OXXO in its area. Personal theft rates are calculated only for the 2015–2018 period, and sexual violence rates only for 2015–2021, as I was unable to obtain SIEDCO data beyond 2018 and 2021, respectively. Always treated units are excluded from the analysis.

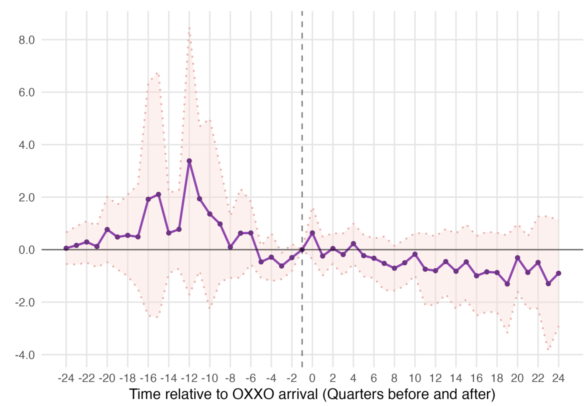
Figure 7 shows the ES of each crime type for the TWFE approach. Neither personal theft (panel 7a) nor sexual offenses (panel 7b) meet the parallel trends assumption near period -1, meaning that their estimates are not correctly identified with this empirical strategy. On the other hand, panels 7c, 7d, and 7e do not show this problem, and after the first OXXO arrival they maintain statistically insignificant dynamic treatment effects, which align with the insignificance of Table 3 coefficients.

Personal theft and sexual violence are the only crime categories that exhibit a temporal trend after the opening of the first OXXO store. Although their patterns are consistent with the sign of their TWFE coefficient and could suggest that OXXO increases personal theft and slightly decreases sexual violence in the medium- and long-term, such interpretations would only hold if the parallel trends were satisfied. Since this condition is not met, it is more likely that the coefficients are biased and incorrectly identified.

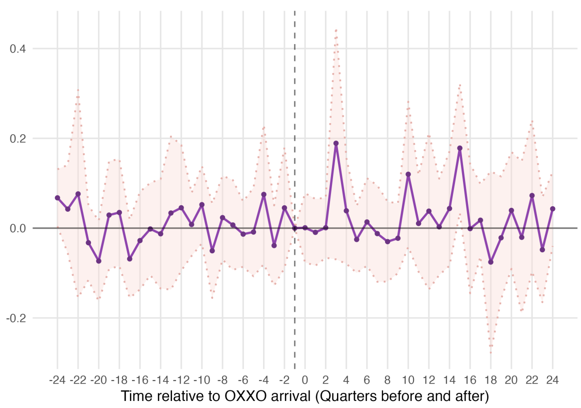
As for heterogeneous treatment effects, both Goodman-Bacon and C&S show that the TWFE regressions do not present this bias. The former decomposition obtained positive weights, while the latter model displayed similar patterns to the TWFE. Table 4



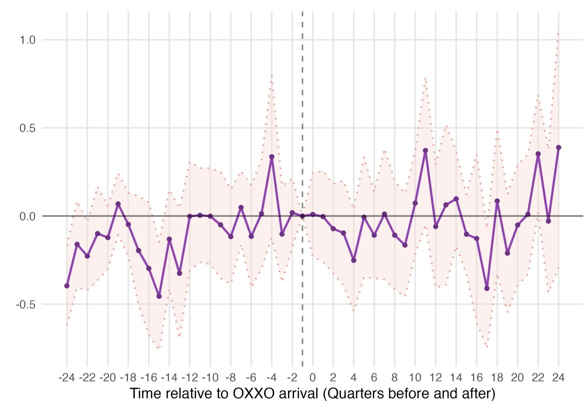
(a) Personal Theft



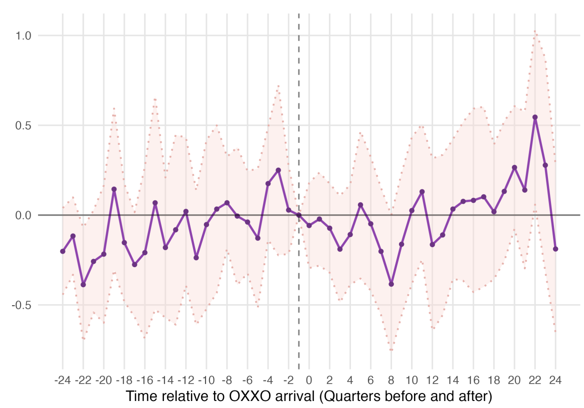
(b) Sexual Violence



(c) Homicide



(d) Motorcycle Theft



(e) Vehicle Theft

Figure 7: Event Study Analysis: TWFE Estimates

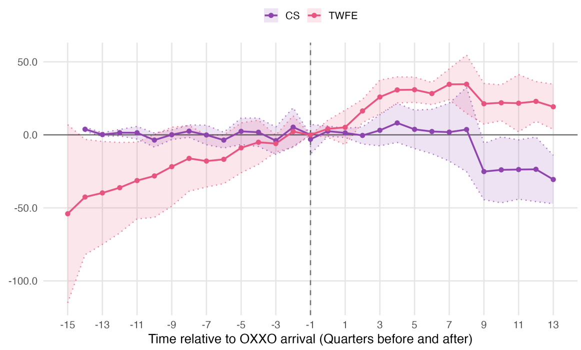
Note: The rate of each crime is per 10,000 inhabitants. I excluded always treated units from the analysis. The dashed vertical line indicates the quarter before the first OXXO opening in each UPZ. The horizontal line at zero represents no effect. The shaded areas represent 95% confidence intervals.

shows the ATT comparison between TWFE and C&S. Homicide and sexual violence requires further examination. In the case of sexual violence, the estimates differ substantially in both magnitude and sign, suggesting the presence of potential bias. As for homicide, both estimators produce coefficients of similar magnitude and direction, yet the estimate becomes statistically insignificant under the C&S specification. Given the consistency of the estimated effects across approaches, the difference appears to stem primarily from a loss of statistical precision due to an increase of the standard error rather than from a substantive change in the estimated impact.

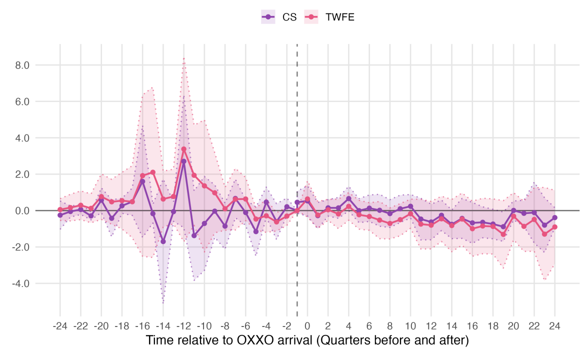
The C&S estimates for personal theft and sexual violence differ from their corresponding TWFE coefficients in both magnitude and sign, while motorcycle theft and vehicle theft retain the same sign but show smaller magnitudes and larger standard errors under C&S. To assess whether these differences reflect genuine bias in the TWFE specification, their respective event studies were compared. Figure 8 suggests that although TWFE specification does not meet the parallel trends assumption in all crimes, after period 0, the event studies of both econometric approaches show almost an equivalent evolution across all quarters. This indicates that, despite the differences, the TWFE specification seemingly presents no bias in terms of heterogeneity.

Another finding depicted in figure 8 is that personal theft's C&S event study evolves in the same direction and similarly to the TWFE specification. Both event studies start showing increasing dynamic treatment effects after period one, and a decreasing trend after quarter seven. Although the C&S estimates eventually become negative, the overall temporal pattern is alike across the two estimators. This consistency shows that if OXXO affects personal theft, the effect is likely strongest in the first post-treatment periods and gradually diminishes over time.

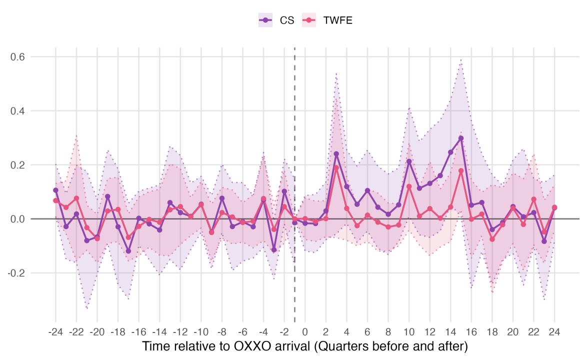
Finally, Table 5 presents the TWFE estimates using treatment intensity. The results are substantially similar to those reported in Table 3 in terms of statistical significance. Personal theft, in particular, keeps exactly the same estimate, because the UPZs treated between 2015 and 2018 have only one store each, which resembles the exact regression as the binary TWFE approach. Consequently, it should be considered that if data after 2018 were to be used, the coefficient would change. All of the remaining crime categories retained the same sign, but are smaller in magnitude compared to those obtained under the binary treatment specification. In this case, the treatment intensity speci-



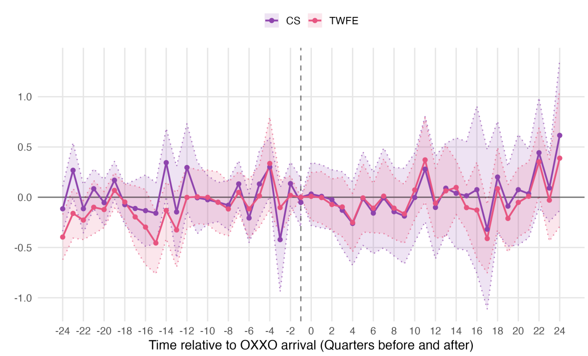
(a) Personal Theft



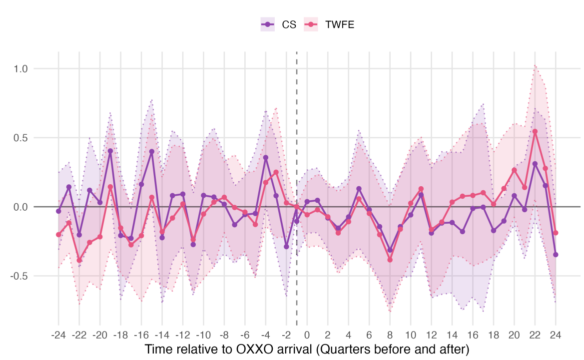
(b) Sexual Violence



(c) Homicide



(d) Motorcycle Theft



(e) Vehicle Thefts

Figure 8: Event Study Analysis: TWFE vs. Callaway & Sant'Anna (C&S) Estimates

Note: The rate of each crime is per 10,000 inhabitants. I excluded always-treated units from both C&S and TWFE Event Studies analysis. The dashed vertical line indicates the quarter before the first OXXO opening in each UPZ. The horizontal line at zero represents no effect. The shaded areas represent 95% confidence intervals.

Table 4: Comparison of ATT Estimates from TWFE and C&S Across Different Crime Types

Crime rates	Personal Theft		Sexual Violence		Homicide		Motorcycle Theft		Vehicle Theft	
Method	TWFE	C&S	TWFE	C&S	TWFE	C&S	TWFE	C&S	TWFE	C&S
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
ATT	2.331	−1.917	−0.440**	0.008	0.084**	0.080	−0.085	−0.029	−0.132	−0.058
SE ATT	(7.141)	(6.699)	(0.199)	(0.224)	(0.033)	(0.070)	(0.080)	(0.165)	(0.087)	(0.119)
Observations	1,440	1,440	2,520	2,520	2,880	2,880	2,880	2,880	2,880	2,880

Note: Standard errors in parentheses. *** p<0.01, ** p<0.05, * p<0.1. ATT is the Average Treatment Effect on the Treated. For TWFE, this is the coefficient presented in Table 3 and for C&S is the weighted average of the ATTs by cohort. Overall, the analysis covers 32 quarters for vehicle theft, motorcycle theft, and homicide, 28 quarters for sexual violence, and 16 quarters for personal theft, reflecting the availability of data for each crime category. Always treated units are excluded from the analysis.

fication estimates the marginal effect of an additional OXXO store; consequently, the smaller coefficients obtained under the intensity specification should not be interpreted as weaker effects, but rather as the average contribution of each additional store to the total crime rate.

Table 5: Effects of OXXO Store Openings on Different Crime Rates under the Treatment-Intensity Specification

Crime rates	Personal theft (1)	Sexual Violence (2)	Homicide (3)	Motorcycle Theft (4)	Vehicle Theft (5)
Number of OXXO stores	2.331	-0.297^{**}	0.035^{**}	-0.014	-0.023
	(7.141)	(0.115)	(0.016)	(0.029)	(0.034)
Constant	30.96^{***}	2.194^{***}	0.464^{***}	1.514^{***}	1.355^{***}
	(0.263)	(0.018)	(0.004)	(0.007)	(0.008)
Observations	1,440	2,520	2,880	2,880	2,880
Treated (UPZ)	9	30	40	40	40
Never treated	81	60	50	50	50
R-squared	0.660	0.417	0.395	0.610	0.582
Mean never treated	55.531	2.167	0.495	1.643	1.153

Note: Standard errors in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.
TWFE considers fixed effects by unit and time, with standard errors clustered by UPZ. The dependent variable is the crime type rate per 10,000 inhabitants per UPZ. Moreover, the treatment variable is continuous and indicates the number of OXXO stores in the UPZ. Personal theft rates are calculated only for the 2015–2018 period, and sexual violence rates only for 2015–2021, as I was unable to obtain SIEDCO data beyond 2018 and 2021, respectively. Always treated units are excluded from the analysis.

Incorporating treatment intensity constitutes a relevant extension of the baseline analysis, given the heterogeneous evolution in the number of OXXO stores across treatment cohorts illustrated in Figure 5. The intensity specification better captures differences in exposure to OXXO presence that are not reflected by a binary treatment indicator. Since the C&S estimator is not defined for continuous treatment intensity, these estimates cannot be directly validated using this approach.

9 Robustness

To assess the sensitivity of the results, I conducted two robustness checks. First, I incorporated the presence of hard-discount store chains, namely Justo & Bueno, D1, and Ara, as controls in all specifications. This considers the potential confounding effect of competing hard-discount retailers, which may simultaneously influence

OXXO location decisions and local criminal activity, thereby biasing the estimates obtained. Second, following Chalfin and McCrary (2018) and Raphael and Winter-Ebmer (2001), I re-estimated all regressions using $log(1 + crime\_type\_index_{upz,quarter})$, where $crime\_type\_index_{upz,quarter}$ coresponds to the usual crime index per 10,000 inhabitants defined in equation 1 [5]. This logarithmic specification reduces the influence of outlier values arising from low residential populations and unusual high crime counts in some UPZs, and, as Pina-Sánchez et al. (2022) argue, could mitigate the variance inflation created by the measurement error in crime data, a particularly relevant concern given that Colombian administrative crime records are subject to systematic underreporting. In both robustness checks I excuded always-treated units from the estimation sample to maintain the same observations used in the Results section.

9.1 Hard discount stores as control variables

The inclusion of these variables did not substantially alter the main observations. I first incorporated D1, Ara, and Justo & Bueno as dummies in the TWFE and C&S models and subsequently as continuous variables in the treatment intensity specification. Overall, the estimated effects remained with the same statistical significance across nearly all crime categories. The only exceptions were vehicle theft and sexual violence in the TWFE dummy specification (found in Table E.1). Vehicle theft became statistically significant and exhibited a larger coefficient, suggesting that previous results underestimated its effect. In contrast, sexual violence lost statistical significance and its estimated effect decreased in magnitude, implying the opposite. Nonetheless, these changes do not appear to be convincing, as the treatment intensity specification (Table E.3) yielded virtually identical coefficients and levels of statistical significance to those reported in Table 5. Taken together, this suggests that competition from hard-discount stores does not meaningfully confound the estimated relationship between OXXO and crime, and that UPZ fixed effects already absorb most of the spatial overlap between these retail chains documented in Appendix C.

Regarding the event studies comparison, the violation of the parallel trends assumption became less pronounced. All crimes, except personal theft and sexual violence,

[5] I added one to the crime rate before applying the log transformation to avoid dropping UPZs with a crime rate of zero. The share of observations with zero crime rates varies across crime types: 2.7% for personal theft, 12.6% for vehicle theft, 11.9% for motorcycle theft, 6.3% for sexual violence, and 32.5% for homicide.

maintained a similar behavioral pattern across quarters. For personal theft, the increasing dynamic treatment effects that were previously observed in figure 8 were attenuated, and after period seven the ES of each model diverged: one (C&S) conserved an increasing dynamic, while the other (TWFE) presented a decreasing tendency. These results indicate that, after considering hard discount stores, none of the crimes analyzed have a temporal tendency following OXXO's entry into a UPZ. The complete set of estimates and ES is reported in Appendix E.

9.2 Transformation of the dependent variable

There are two reasons why the estimates obtained may not show the real impact of OXXO presence. First, the measurement error and its effects on precision in the dependent variables could be the reasons why almost all crimes ended with statistically insignificant estimates. Second, a number of UPZs reported high crime counts relative to their residential population, even after excluding always treated units. Thus, the log transformation is expected to mitigate these two issues, and provide a diagnostic test of whether these mechanisms were affecting the original estimates.[6]

Table 6 reports the estimates obtained after applying the log transformation and compares the regressions with and without hard-discount store controls. Unlike the baseline specification, the coefficients for personal theft, homicide, and motorcycle theft (columns (1), (5), and (7)) are now statistically significant. Additionally, vehicle theft (column (10)) remains statistically significant, although only at the 10% level, consistent with the previous robustness check that included hard-discount store controls. The specification which includes these controls indicates that UPZs with OXXO presence were, in almost all cases, underestimated. Additionally, regressions present a 15.7% increase in personal theft, a 5.9% increase in homicides, an 8.9% reduction in motorcycle theft, a 16% reduction in sexual violence, and an 8.3% reduction in vehicle theft compared to UPZs without OXXO stores.

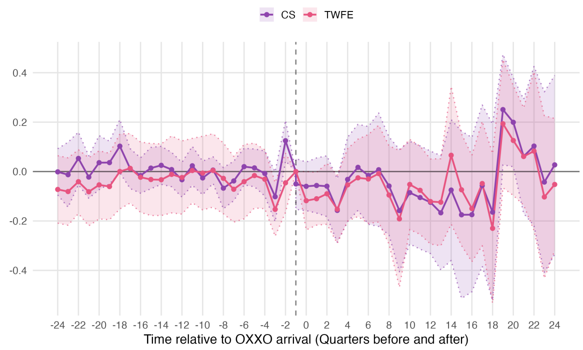
These findings suggest that the transformation was effective in mitigating both out-

[6] It is worth noting that the log transformation only addresses measurement error when it is multiplicative $U*Y$ in stead of $U+Y$, only under this condition does the transformation convert proportional noise into a more tractable additive form on the log scale (Pina-Sánchez et al., 2022; Wooldridge, 2010). There is no formal test to determine whether measurement error is multiplicative or additive, but an informal indicators could be whether the estimates' presicion improves.

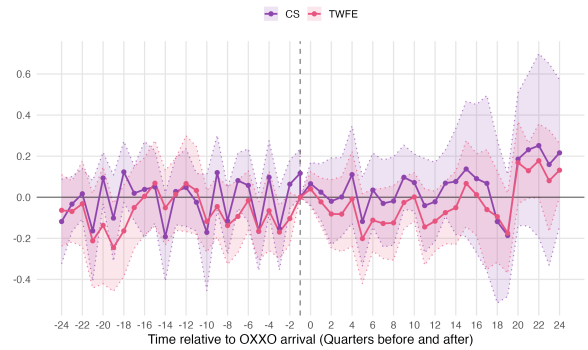
liers and the imprecision caused by measurement error (although it does not eliminate the possible bias arising from it), without evidencing inconsistent results. In particular, comparing the standard errors relative to their coefficients (i.e., the t-statistics) across the levels (Table E.1) and log-transformed (Table 6) specifications, I find that personal theft and motorcycle theft experience the largest gains in precision. The t-statistic for personal theft increases from 0.33 to 2.96, while that for motorcycle theft rises in absolute value from -1.06 to -2.17. Vehicle theft shows a smaller but still meaningful gain, from -1.52 to -1.77. Sexual violence, already significant in levels, becomes even more precisely estimated (from -2.21 to -3.14). Homicide is the only category that remains virtually unchanged (2.47 to 2.36), consistent with the expectation that homicide, being the crime least subject to underreporting, has comparatively little measurement error for the transformation to correct.

This could indicate that the original specification understated the effects of OXXO on some crime categories. However, the results reported in Table 7 challenge this interpretation. Even with the log specification, all C&S estimates remained statistically insignificant. Although this pattern is consistent across all robustness checks and the baseline analysis, the lack of statistical significance in the C&S estimates may be partly attributable to the way the estimator is constructed. C&S first computes separate $\text{ATT}(g, t)$ estimates for each treatment cohort (Equation 4) and then aggregates these into a weighted average, which can increase the standard errors and, consequently, reduce the statistical significance of the coefficients. Additionally, C&S is primarily used as a diagnostic test for treatment effect heterogeneity in TWFE estimates, and the results presented above provide no evidence of such bias in the TWFE specification.

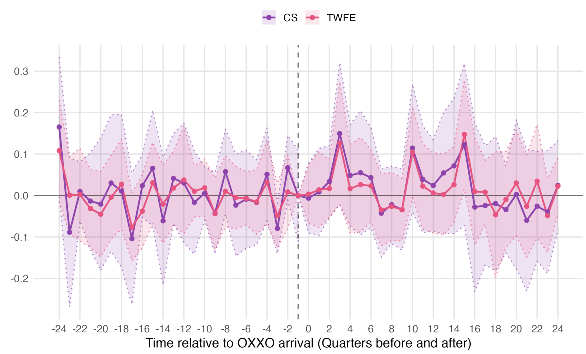
Finally, the ES for both models (without controls) maintained the same dynamic treatment effects as the baseline results. Only personal theft and sexual violence, as presented in panels 9a and 9b of Figure 9, mirror the behavior observed when hard-discount stores were added. Specifically for personal theft, it no longer exhibits the increasing dynamic treatment effects that previously characterized this crime. This strengthens the conclusion that, in fact, OXXO had no medium- to long-term effects on personal theft, and that this behavior was driven by UPZ outliers and other variables omitted in the simpler regressions.



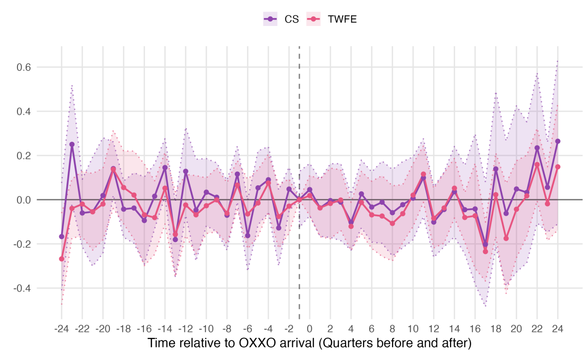
(a) Personal Theft



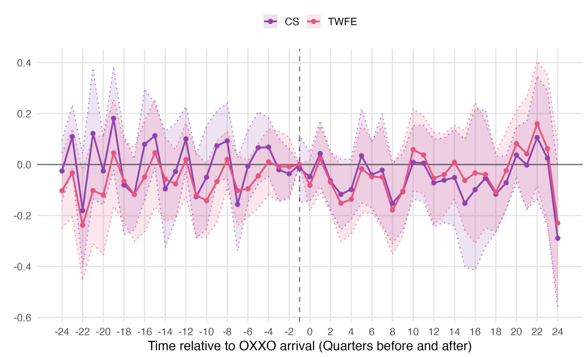
(b) Sexual Violence



(c) Homicide



(d) Motorcycle Theft



(e) Vehicle Theft

Figure 9: Event Study Analysis: TWFE Estimates

Note: The rate of each crime number is the logarithm of the smoothed crime type rate per 10,000 inhabitants per UPZ. Always-treated units are excluded from both the C&S and TWFE Event Study analyses. The dashed vertical line indicates the quarter before the first OXXO opening in each UPZ. The horizontal line at zero represents no effect. The shaded areas represent 95% confidence intervals.

Table 6: Effects of OXXO Store Openings on Different Crime Rates under the log transformation of the dependent variable

Crime rates	Personal Theft		Sexual Violence		Homicide		Motorcycle Theft		Vehicle Theft	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
Has OXXO	0.160^{***}	0.157^{***}	-0.123^{**}	-0.160^{***}	0.057^{**}	0.059^{**}	-0.077^{*}	-0.089^{**}	-0.050	-0.083^{*}
	(0.051)	(0.053)	(0.059)	(0.051)	(0.027)	(0.025)	(0.039)	(0.041)	(0.043)	(0.047)
Constant	2.837^{***}	2.813^{***}	0.979^{***}	1.019^{***}	0.317^{***}	0.315^{***}	0.812^{***}	0.788^{***}	-0.050	0.701^{***}
	(0.002)	(0.015)	(0.007)	(0.022)	(0.004)	(0.011)	(0.006)	(0.016)	(0.043)	(0.013)
Chain Stores	no	yes	no	yes	no	yes	no	yes	no	yes
R-squared	0.930	0.909	0.473	0.476	0.491	0.426	0.500	0.501	0.505	0.500
Observations	1,440		2,520		2,880		2,880		2,880	
Treated (UPZ)	9		30		40		40		40	
Never Treated	81		60		50		50		50	
Mean Never Treated	55.531		2.167		0.495		1.643		1.153	

Note: Standard errors in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.
TWFE considers fixed effects by unit and time, with standard errors clustered by UPZ. The dependent variable is the logarithm of the smoothed crime type rate per 10,000 inhabitants per UPZ. Moreover, the treatment variable is binary and indicates if the UPZ has at least one OXXO in its area. Personal theft rates are calculated only for the 2015–2018 period, and sexual violence rates only for 2015–2021, as I was unable to obtain SIEDCO data beyond 2018 and 2021, respectively. Hard-discount controls are included as dummies.

Table 7: Comparison of ATT Estimates from TWFE and C&S Across Different Crime Types under the log transformation of the dependent variable

Crime rates	Personal Theft		Sexual Violence		Homicide		Motorcycle Theft		Vehicle Theft	
Method	TWFE	C&S	TWFE	C&S	TWFE	C&S	TWFE	C&S	TWFE	C&S
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
ATT	0.160^{***}	0.063	-0.123^{**}	-0.010	0.057^{**}	0.030	-0.077^{*}	-0.011	-0.050	-0.051
SE ATT	(0.051)	(0.065)	(0.059)	(0.076)	(0.027)	(0.041)	(0.039)	(0.060)	(0.043)	(0.040)
Observations	1,440	1,440	2,520	2,520	2,880	2,880	2,880	2,880	2,880	2,880

Note: Standard errors in parentheses. *** $p<0.01$, ** $p<0.05$, * $p<0.1$. ATT is the Average Treatment Effect on the Treated. For TWFE, this is the coefficient presented in Table 6 and for C&S is the weighted average of the ATTs by cohort. Overall, the analysis covers 32 quarters for vehicle theft, motorcycle theft, and homicide, 28 quarters for sexual violence, and 16 quarters for personal theft, reflecting the availability of data for each crime category. Always treated units are excluded from the analysis.

10 Discussion

Overall, although the C&S estimates do not evidence that the expansion of OXXO stores over time affected most of the crime categories analyzed, I place greater weight on the statistically significant TWFE estimates obtained in the robustness checks. Both models had similar coefficients in Table 7 and the TWFE evidenced no heterogeneous treatment effect bias, the main difference was that C&S estimates were smaller in magnitude and had greater standard errors, which was a result of differences in how each estimator is constructed, rather than any underlying bias.[7]

The only crimes that remained statistically significant across all specifications were homicide and sexual violence. In contrast, personal theft, vehicle theft, and motorcycle theft became statistically significant only after applying the log transformation. Although I expected these last two crimes to exhibit the same direction of effect as personal theft, there are several mechanisms that may explain why vehicle and motorcycle theft decline following the entry of an OXXO store. In the case of personal theft, studies such as Askey et al. (2018), Brantingham and Brantingham (1995), and Dugato et al. (2025) reported that similar establishments were attractors for this type of crime as they increase the concentration of potential victims in a single spatial point. However, this same mechanism (namely, the higher concentration of people or increased pedestrian activity) may help explain the reduction in motorcycle and vehicle theft. Unlike personal theft, these two crimes typically require more time and greater effort to be committed without being detected. Consequently, greater pedestrian activity may enhance natural surveillance by providing more *eyes on the street* (Jacobs, 1961), thereby increasing the likelihood that offenders are observed and deterring the theft of unattended vehicles (Jaramillo López et al., 2021; Kefayat & Thill, 2025). Additionally, OXXO's integration of delivery services (OXXO Colombia, 2026b) may further increase surveillance of parked vehicles, as delivery personnel tend to monitor their motorcycles and bicycles while entering and leaving the store (Zhou et al., 2025).

[7]This difference in precision stems from how each estimator aggregates information. TWFE pools all treated and control observations into a single regression. The C&S estimator, by contrast, first computes separate $\text{ATT}(g, t)$ estimates for each treatment cohort (Equation 4), each based on a smaller subset of the data, and then aggregates these cohort-specific estimates into a weighted average (Equation 5). Because each cohort-period ATT is estimated with fewer observations than the pooled TWFE regression, and because the aggregation step introduces additional estimation uncertainty on top of each individual $\text{ATT}(g, t)$, the resulting standard errors for the aggregate C&S estimate tend to be larger, even when the underlying point estimates are similar.

In terms of dynamic treatment effects, the event studies demonstrate that, across all analyzed crime categories, the impacts of OXXO establishments do not appear to increase or decrease over the medium or long run. Personal theft may have been the only one that presented a pronounced tendecy in the baseline results. However, robustness checks did not confirm this evolution; while they corrected the parallel trends violation present in the baseline model, they also removed the statistical significance of the dynamic treatment effect. Contrary to this finding, Jaramillo López et al. (2021) and Shin (2024) reported diverging dynamic effects on theft for similar commercial establishments finding a post-treatment decrease and increase, respectively. These differences suggest that the results may differ across contexts and provide further support for the statistically insignificant dynamic treatment effects, which are consistent with the C&S estimates.

The null effect of most crimes in the baseline specification can be can be attributed to several underlying factors. One explanation is that the measurement error of crime rates is inflating the standard errors, thereby rendering the estimations statistically insignificant. This conjecture is reinforced by the fact that homicide is one of the two statistically robust crimes, a crime that hardly presents any measurement error as it is difficult to underreport. Furthermore, after the log transformation in Section 9.2 almost all crimes resulted significant, a finding that strengthens this hypothesis. Another explanation is that, after all, OXXO is not a relevant actor in shaping the patterns of criminal activity in Bogotá. Other studies, such as Askey et al. (2018), Davis et al. (2021), and Feng et al. (2019), evaluate commercial establishments collectively rather than isolating a single corporate chain; this broader treatment definition may capture a stronger and more statistically detectable aggregate effect than my single-chain analysis.

Homicide and sexual violence are the only crime categories that remain robustly significant across the specifications. For sexual violence I did not initially expect a significant effect from OXXO's arrival. Since this crime is often assumed to occur predominantly in secluded, indoor, or otherwise private settings, I intended to use it as a placebo outcome to assess whether the model was producing spurious results. Nonetheless, upon closer inspection of the *victim mobility modality* recorded in the raw SIEDCO data, I found that 99.78% of cases involved a victim who was walking at the time of the incident. This makes the regression results more coherent, as the estimated effect is localized: OXXO presence is associated with a 16% reduction in sexual violence incidents occurring on the street, and incidents linked to street-level exposure are precisely those that OXXO's presence could most plausibly influence. A possible explanation is the same mechanism proposed for vehicle and motorcycle theft: increased *eyes on the street* during both daytime and nighttime hours.

Conversely, the effect found in homicide can be explained through the results of other similar studies. Empirical evidence from Scribner et al. (1999) and Bowen et al. (2022) provides a plausible mechanism for why OXXO openings alter homicide dynamics in Bogotá. Scribner et al. (1999), for instance, studied the effect of alcohol outlets on homicide in New Orleans, finding a positive relationship and an increase in violence in areas with higher establishment density. Moreover, Bowen et al. (2022) corroborated these findings for convenience stores, and commented that 24-hour retail operations more than doubled the odds of violent injuries in their surroundings compared to stores operating under restricted hours. Although OXXO stores are not exclusively alcohol outlets, they sell alcoholic beverages and operate under a 24-hour system. Under these distinct operational characteristics, and from a public safety perspective, OXXO establishments can inadvertently act as ambient crime generators (Davis et al., 2021). By functioning as highly visible, illuminated nodes, they can become focal points for nocturnal gatherings, facilitate alcohol consumption in adjacent public spaces or within the same street, and provide the conditions for spontaneous disputes. Moreover, these dynamics can rapidly escalate into lethal altercations or, as my empirical findings suggest, homicides. For subsequent analyses of this thesis, it will be important to explore whether the statistical significance and magnitude of the homicide effect is driven by nocturnal homicides. If true, this will further reinforce the idea that OXXO may be a crime generator during nighttime hours.

Finally, it is important to emphasize that the mechanisms proposed here are only plausible explanations and may not represent the actual channels through which OXXO affects crime. Moreover, the estimated effects may still be subject to bias. For the mechanisms described, additional tests should be run to determine whether they are indeed responsible for the observed increases or decreases in crime. For example, to assess whether alcohol plays a relevant role in increasing homicide rates, one approach would be to estimate the same specifications for a similar establishment such as *Farmatodo*, a convenience store that runs 24 hours but does not sell alcholic beverages. If no effect on homicide is found for *Farmatodo*, this would provide suggestive evidence that alcohol sales contribute to OXXO's impact on this crime. On the other hand, bias in the estimated coefficients may still be a concern, as the log transformation does not eliminate measurement error but only reduces the imprecision it introduces. Consequently, the estimated effects may still be attenuated. Additionally, I did not account for the possibility of spillover effects from treated UPZs to nearby control units. OXXO stores located near control UPZs could displace crime toward those areas, thereby biasing the estimated treatment effects.

11 Conclusions

This study exploits TWFE and C&S specifications to show how the staggered arrival of OXXO convenience stores affect local crime activity in Bogotá's Zonal Planning Units, and whether effects vary by crime type and time since opening. At first, my baseline specification found that homicide and sexual violence were the only crimes that were shaped by OXXO arrival; nonethelss, the log transformation in crime rates plus the inclusion of hard-discount stores as covariates evidenced that, in fact, all crimes evaluated (at different levels of significance) were affected by OXXO, but in a different direction. Theft to people and homicide increased, while sexual violence, and vehicle and motorcycle theft decreased. These results may be driven by increased pedestrian activity and nocturnal gatherings, which creates opportunities for personal theft and homicide while simultaneously enhancing natural surveillance and reducing the other crimes.

These estimates are consistent with previous research, which show that personal theft and homicides rates increase under the presence of similar establishments (Askey et al., 2018; Bowen et al., 2022; Brantingham & Brantingham, 1995; Dugato et al., 2025). The main difference is that prior studies do not typically report opposing effects across crime categories, suggesting that results are tied to specifical attributes of each city. Additionally, these findings are important from both a substantive and an econometric perspective. On one hand, they help to explore in greater depth the urban dynamics of Latin American cities, as studies of this nature tend to focus on high-income contries´ urban areas, which have different characteristics and even convenience stores tend to locate in places with distinct urban and economical backgrounds. On the other, they verify that the log transformation may be an effective strategy to reduce variance inflation due to measurement error; this specification is often overlooked for this reason and tends to be used to correct positively skewed data (Pina-Sánchez et al., 2022).

This research has important implications for urban policy and crime prevention strategies. Results provide a broader understanding of the additional urban dynamics that may shape patterns of criminal activity in Bogotá, as well as the role that commercial establishments play in these dynamics. Futhermore, this suggests an opportunity for stronger public-private coordination. Given that FEMSA centrally administers store locations and operating hours, collaboration with security authorities may be easier to coordinate, allowing for joint monitoring agreements aimed at disrupting the mechanisms through which these crimes occur. At the end, although the expansion of these establishments may increase the risk of homicide and personal theft, at the same time, OXXO stores may also generate positive externalities, such as reductions in vehicles theft and sexual violence.

Finally, this study has several limitations that warrant careful consideration. The first is the unit of analysis I chose. Aggregating variables at the UPZ level may introduce measurement challenges, as it treats crimes committed far from a store identically to those occurring in its immediate vicinity. In contrast to this approach, other empirical literature uses localized areas by employing narrow distance buffers around each establishment to define treated and control units (Jaramillo López et al., 2021; Shin, 2024). This approach is also more effective at mitigating spillover effects, since overlapping areas can be omitted from the estimation while retaining a sufficiently large number of observations. This unit specification consideration could be explored in future research to determine whether focusing only on the immediate vicinity yields similar results. The second limitation concerns the limited time coverage of the personal theft data. The results for this crime category should therefore be interpreted with caution, as the short study period (2015–2018) provides little variation in treatment status across units. Finally, the third limitation is that this study does not evaluate whether the effect of OXXO differs statistically between crimes committed during the day versus at night. Future research can explore this distinction, particularly for homicide and personal theft, as time-of-day disaggregation would provide a direct test of the mechanisms I proposed in this thesis.

References

Alcaldía Mayor de Bogotá. (2021). ¿qué son las Unidades de Planeamiento Zonal (UPZ)? [Accessed: 2026-04-01]. https://bogota.gov.co/mi-ciudad/salud/que-es-una-upz

Askey, A. P., Taylor, R., Groff, E., & Fingerhut, A. (2018). Fast food restaurants and convenience stores: Using sales volume to explain crime patterns in Seattle. *Crime & Delinquency*, *64*(14), 1836–1857. https://doi.org/10.1177/0011128717714792

Bacares, O., & Augusto, C. (2013). *Criminality spread: A "boomerang effect" of public transport improvements?* (Tech. rep.). Palermo, Italy, Louvain-la-Neuve.

Bacares, O., & Augusto, C. (2014). *Impact of urban public transport enhancements on crime rate: A diff-diff analysis for the case of Transmilenio* (Working Paper). Research Papers in Economics. https://ideas.repec.org/p/col/000089/011116.html

Blattman, C., Green, D. P., Ortega, D., & Tobón, S. (2017). *Pushing crime around the corner? estimating experimental impacts of large-scale security interventions* (NBER Working Paper No. 23035). National Bureau of Economic Research. https://doi.org/10.3386/w23035

Blattman, C., Green, D. P., Ortega, D., & Tobón, S. (2021). Place-based interventions at scale: The direct and spillover effects of policing and city services on crime. *Journal of the European Economic Association*, *19*(4), 2022–2051. https://doi.org/10.1093/jeea/jvab002

Bocarejo, J. P., & Tafur, L. E. (2013). *Urban land use transformation driven by an innovative transportation project, Bogotá, Colombia* (tech. rep.) (Case study prepared for the Global Report on Human Settlements 2013). United Nations Human Settlements Programme (UN-Habitat). http://www.unhabitat.org/grhs/2013

Bowen, D. A., Anthony, K. M., & Sumner, S. A. (2022). Business and property types experiencing excess violent crime: A micro-spatial analysis [Epub 2021 Nov 17]. *Journal of Injury and Violence Research*, *14*(1), 1–10. https://doi.org/10.5249/jivr.v14i1.1566

Brantingham, P., & Brantingham, P. (1995). Criminality of place: Crime generators and crime attractors. *European Journal on Criminal Policy and Research*, *3*(3), 5–26. https://doi.org/10.1007/BF02242925

Callaway, B., & Sant'Anna, P. H. C. (2021). Difference-in-Differences with multiple time periods. *Journal of Econometrics*, *225*(2), 200–230. https://doi.org/10.1016/j.jeconom.2020.12.001

Cámara de Comercio de Bogotá. (2025). *Decreasing perception of insecurity and victimization in Bogotá*. Retrieved May 10, 2026, from https://www.ccb.org.co/en/of-interest/communications/decreasing-perception-of-insecurity-and-victimization-in-bogota

Cámara de Comercio de Bogotá. (2026a). Evolución de las empresas activas por localidad de Bogotá [Accessed: 2026-07-22]. *Cámara de Comercio de Bogotá*.

Cámara de Comercio de Bogotá. (2026b, February). *Bogotá mejoró en indicadores de victimización y percepción de seguridad en 2025* [Consultado el 31 de marzo de 2026]. https://www.ccb.org.co/de-interes/comunicados/bogota-mejoro-en-indicadores-de-victimizacion-y-percepcion-de-seguridad-en-2025

Chalfin, A., & McCrary, J. (2018). Are U.S. cities underpoliced? Theory and evidence. *The Review of Economics and Statistics*, *100*(1), 167–186. https://doi.org/10.1162/REST_a_00694

Clarke, R. V., & Mayhew, P. (1994). Parking patterns and car theft risks: Policy-relevant findings from the British Crime Survey. *Crime Prevention Studies*, *3*, 91–107.

D'Alessio, S. J., & Stolzenberg, L. (1990). A crime of convenience: The environment and convenience store robbery. *Environment and Behavior*, *22*(2), 255–271. https://doi.org/10.1177/0013916590222004

Davis, J., Casteel, C., & Chaumont Menéndez, C. (2021). Impact of a crime prevention ordinance for small retail establishments. *American Journal of Industrial Medicine*, *64*(6), 488–495. https://doi.org/10.1002/ajim.23239

de Chaisemartin, C., & D'Haultfœuille, X. (2023). Two-way fixed effects and differences-in-differences with heterogeneous treatment effects: A survey. *The Econometrics Journal*, *26*(3), C1–C30. https://doi.org/10.1093/ectj/utac017

Delgado, L., Otero, A., & Calderón, A. (2024). *The impact of hard discount stores on local labor markets: Evidence from Colombia* (tech. rep. No. 326). Centro De Estudios Económicos Regionales - Cartagena. https://www.banrep.gov.co/en/documentos-trabajo-sobre-economia-regional-urbana-impact-hard-discount-stores-local-labor-markets

de Mello, J., Mejía, D., & Suárez, L. (2013). *The pharmacological channel revisited: Alcohol sales restrictions and crime in Bogotá* (Working Paper No. 2262335). Social Science Research Network (SSRN). https://doi.org/10.2139/ssrn.2262335

Domínguez, P., & Asahi, K. (2022). Crime-time: How ambient light affects crime. *Journal of the European Economic Association*, *20*(6), 2457–2498. https://doi.org/10.1093/jeg/lbac011

Dugato, M., Giménez-Santana, A., Santos, A., Caplan, J. M., & Kennedy, L. W. (2025). "not all convenience stores are equal": An analysis of the criminogenic role of corner stores in Newark, NJ. *Journal of Criminal Justice*, *102*, 102552. https://doi.org/10.1016/j.jcrimjus.2025.102552

El Economista. (2026). Oxxo se expande en colombia: Abre una tienda cada tres días. *El Economista*. https://www.eleconomista.com.mx/empresas/oxxo-expande-colombia-abre-tienda-tres-dias-20260416-808982.html

Enriquez, D. F. (2014). Neighborhood characteristics and the spatial concentration of crime: Evidence for Bogotá. *Universidad del Rosario*, *74*, 131–170. https://doi.org/10.13043/DYS.74.4

Escobar, G. A. (2012). El uso de la teoría de la desorganización social para comprender la distribución de homicidios en Bogotá, Colombia. *Revista INVI*, *27*(74), 21–85. https://doi.org/10.4067/S0718-83582012000100002

Feng, S. Q., Piza, E. L., Kennedy, L. W., & Caplan, J. M. (2019). Aggravating effects of alcohol outlet types on street robbery and aggravated assault in New York City. *Journal of Crime and Justice*, *42*(3), 257–273. https://doi.org/10.1080/0735648X.2018.1559076

García, J. (2016, May). Inducción Alcaldes Locales: Seguridad Local – Plan Integral de Seguridad, Convivencia y Justicia (PISCJ) [Accessed: 2026-07-21]. Retrieved July 21, 2026, from https://historico.gobiernobogota.gov.co/sites/gobiernobogota.gov.co/files/documentos/tabla_archivos/10.seguridad_local_-_pics_06-05-16_dr._jairo_garcia.pdf

Giménez-Santana, A., Caplan, J. M., & Drawve, G. (2018). Risk terrain modeling and socio-economic stratification: Identifying risky places for violent crime victimization in Bogotá, Colombia. *European Journal on Criminal Policy and Research*, *24*(4), 457–476. https://doi.org/10.1007/S10610-018-9374-5

Godoy, J. F., Rodríguez, C., & Zuleta, H. (2018). *Security and sustainable development in Bogotá, Colombia: A case study for the DCAF policy and research division project: SSR for Safer Cities – Supporting States to Achieve SDG 11*. Center of Studies on Security and Drugs (CESED). Bogotá, Colombia, Geneva, Switzerland, DCAF Policy and Research Division.

Godoy, M. C. P. (2025, March 17). *Tiendas oxxo abre su local número 600 en Colombia.* https://www.valoraanalitik.com/tiendas-oxxo-abre-su-local-numero-600-en-colombia/

Goodman-Bacon, A. (2021). Difference-in-differences with variation in treatment timing. *Journal of Econometrics*, *225*(2), 254–277. https://doi.org/10.1016/j.jeconom.2021.03.014

Guzman, L., & Bocarejo, J. P. (2017). Urban form and spatial urban equity in Bogota, Colombia. *Transportation Research Procedia*, *25*, 4495–4510. https://doi.org/10.1016/j.trpro.2017.05.345

Jacobs, J. (1961). *The death and life of great american cities*. Random House.

Jaramillo López, M., Jiménez Quitian, I. F., & Pérez Cepeda, J. P. (2021). *Low-cost, lower crime?: Impacto de los supermercados de bajo costo sobre la criminalidad en Bogotá. El caso de Tiendas D1* [Tesis de Maestría]. Universidad de los Andes. https://doi.org/10.57784/1992/55764

Kefayat, E., & Thill, J.-C. (2025). Urban street network configuration and property crime: An empirical multivariate case study. *ISPRS International Journal of Geo-Information*, *14*(5), 200. https://doi.org/10.3390/ijgi14050200

Konishi, H. (2005). Concentration of competing retail stores. *Journal of Urban Economics*, *58*(3), 88–512. https://doi.org/10.1016/j.jue.2005.08.005

López, M. C. (2026). Entrevista con andrés morales, director general de oxxo colombia [Entrevista sobre el plan de expansión en el mercado colombiano]. *Diario La República*. https://www.larepublica.co/empresas/entrevista-con-andres-morales-director-general-de-oxxo-colombia-4334424

Marcos, M. Á. (2022). Preliminary evidence of surviving competition: Neighborhood shops vs. convenience chains. *Cuadernos de Economía*. https://doi.org/10.18235/0004453

Mockus, A. (2012, April). Building "Citizenship Culture" in Bogotá [Accessed: 2026-04-01]. https://jia.sipa.columbia.edu/news/building-citizenship-culture-bogota

Moreno, Á. J. (2005). *Impacto de Transmilenio en el crimen de la Avenida Caracas y sus vecindades* (Working Paper). Centro de Estudios sobre Desarrollo Económico (CEDE).

Moreno García, A. J. (2005). *Impacto de Transmilenio en el crimen de la Avenida Caracas y sus vecindades* (Documento CEDE No. 2005-55). Universidad de los Andes, CEDE. https://doi.org/10.57784/1992/7963

Nazzari, M., Calaresu, M., & Triventi, M. (2026). No sales after midnight: Evaluating the impact of a business curfew on drug-related crime in San Francisco's Tenderloin [Accepted: 27 October 2025]. *Security Journal*, *39*(5). https://doi.org/10.1057/s41284-025-00517-w

Observatorio de Desarrollo Económico de Bogotá. (2022, December). *Atlas Económico de Bogotá 2022* (Report) (Accessed: 2026-07-22). Secretaría de Desarrollo Económico de Bogotá. https://observatorio.desarrolloeconomico.gov.co/wp-content/uploads/2025/03/2022-12-30-Atlas_economico_2022.pdf

OXXO Colombia. (2026a). Preguntas frecuentes (faqs) [Accessed: 2026-04-01]. https://oxxo.co/faqs

OXXO Colombia. (2026b). *Términos y condiciones particulares de servicio de compras a domicilio en OXXO Colombia por medio de la OXXO app y página web.* Retrieved July 13, 2026, from https://colombia.oxxodomicilios.com/section/tcHomePurchaseService

Pina-Sánchez, J., Buil-Gil, D., Brunton-Smith, I., & Cernat, A. (2022, June). *The impact of measurement error in models using police recorded crime rates* (Preprint) (University of Leeds). SocArXiv. https://doi.org/10.31235/osf.io/ydf4b

Policía Nacional de Colombia. (2026). *Portal público de indicadores del sistema de información estadístico, delincuencial, contravencional y operativo (siedco).* Policía Nacional de Colombia. Retrieved May 10, 2026, from https://portalsiedco.policia.gov.co:4443/extensions/PortalPublico/index.html#/Indicadores/Delito

PRISM. (2025). Partnership for Research on Informal and Shared Mobility (PRISM) Bogotá Living Lab [Accessed: 2026-05-10].

ProBogotá Región. (2023, October). *Informe Anual de Seguridad 2022: Análisis y Perpectivas para Bogotá* (Annual Report) (Analysis of crime trends and institutional response). Fundación ProBogotá. Bogotá, Colombia. https://www.probogota.org/wp-content/uploads/2023/10/Informe-Anual-de-Seguridad-2022.pdf

ProBogotá Región. (2024, July). Estas son las localidades más seguras e inseguras de Bogotá según el Índice Integral de Seguridad [Accessed: 2026-04-01]. https://www.probogota.org/comunicacion_c/estas-son-las-localidades-mas-seguras-e-inseguras-de-bogota-segun-el-indice-integral-de-seguridad/

Raphael, S., & Winter-Ebmer, R. (2001). Identifying the effect of unemployment on crime. *The Journal of Law and Economics*, *44*(1), 259–283. https://doi.org/10.1086/320275

Rico-Pencue, B. E., Vera-Castillo, J. S., & Bohórquez-Arévalo, L. E. (2021). Impacto social, económico y de mercado como consecuencia del Covid-19 en tiendas de barrio en el sur de Bogotá. *Revista EAN*, (91). https://doi.org/10.21158/01208160.n91.2021.3175

Rios-Avila, F., Callaway, B., & Sant'Anna, P. H. C. (2021, August). csdid: Difference-in-Differences with multiple time periods in Stata [Available upon request or at the Stata Conference 2021 materials].

Rodríguez, S. C. (2017). "La ciudad más insegura del mundo": Homicidio y crimen en Bogotá, 1988-1994 [Language: Spanish]. *Revista Criminalidad*, *59*(2), 49–64. http://www.scielo.org.co/scielo.php?pid=S1794-31082017000200049&script=sci_abstract&tlng=es

Sánchez, F., Espinosa, S., & Rivas, Á. (2003). *¿garrote o zanahoria? factores asociados a la disminución de la violencia homicida y el crimen en Bogotá, 1993-2002* (Documento CEDE No. 2003-05). Centro de Estudios sobre Desarrollo Económico (CEDE). Bogotá, Colombia. https://economia.uniandes.edu.co/publicaciones/cede/documentos-cede

Scribner, R., Cohen, D., Kaplan, S., & Allen, S. H. (1999). Alcohol availability and homicide in New Orleans: Conceptual considerations for small area analysis of the effect of alcohol outlet density. *Journal of Studies on Alcohol*, *60*(3), 310–316. https://doi.org/10.15288/jsa.1999.60.310

Seong, E. Y., Lim, Y., & Choi, C. G. (2022). Why are convenience stores clustered? the reasons behind the clustering of similar shops and the effect of increased competition. *Environment and Planning B: Urban Analytics and City Science*, *49*(3), 834–846. https://doi.org/10.1177/23998083211021870

Shin, E. J. (2024). Are dollar stores magnets for violent crime? Evidence from Chicago. *Urban Affairs Review*, *60*(5), 1540–1576. https://doi.org/10.1177/10780874241226875

Summers, L. (2014). Outdoor serious violence: The role of place. In G. Bruinsma & D. Weisburd (Eds.), *Encyclopedia of criminology and criminal justice*. Springer. https://doi.org/10.1007/978-1-4614-5690-2_299

Wellford, C. F., MacDonald, J., & Weiss, J. C. (1997). *Multistate study of convenience store robberies* (tech. rep.) (With the assistance of Timothy Bynum, Robert Friedmann, Robert McManus, and Anthony Petrosino). Justice Research and Statistics Association. Washington, DC. https://www.jrsa.org/pubs/reports/conv-store-robbery.pdf

Wooldridge, J. M. (2010). *Econometric analysis of cross section and panel data* (2nd). MIT Press.

Xu, C., Chen, X., Liu, L., Lan, M., & Chen, D. (2021). Assessing impacts of new subway stations on urban thefts in the surrounding areas. *ISPRS International Journal of Geo-Information*, *10*(10), 632. https://doi.org/10.3390/ijgi10100632

Zhou, S., Liu, L., Yue, H., Yang, K., Li, J., & Song, G. (2025). Can delivery riders act as informal guardians in crime prevention? *Applied Geography*. https://doi.org/10.1016/j.apgeog.2025.103802

12 Appendix

A Detailed description of variables

Table A.1: Data Sources

Variable	Description	Data Source
Dependent Variables		
Personal theft crime rate	Personal theft rate per 10,000 inhabitants	SIEDCO
Homicide rate	Homicide rate per 10,000 inhabitants	SIEDCO
Sexual violence crime rate	Sexual violence rate per 10,000 inhabitants	SIEDCO
Theft of Motorcycles crime rate	Motorcycle theft rate per 10,000 inhabitants	SIEDCO
Theft of Vehicles crime rate	Vehicle theft rate per 10,000 inhabitants	SIEDCO
Staggered Treatment Variable		
Has OXXO (= 1)	Indicates whether the UPZ has at least one OXXO store in a given quarter	RUES and Google Maps API
Staggered Controls: Hard Discount Stores		
Has D1s (= 1)	Indicates whether the UPZ had at least one D1 store in the previous quarter	RUES and Google Maps API
Has ARAs (= 1)	Indicates whether the UPZ had at least one ARA store in the previous quarter	RUES and Google Maps API
Baseline Variables		
Inhabitants by locality in 2007	Number of inhabitants in each locality in 2007	Survey of Quality of Life in Bogotá 2007

Table A.1 – Continued from previous page

Variable	Description	Data Source
Household size 2007	Average number of people per household in each Locality in 2007	Survey of Quality of Life in Bogotá 2007
Index of quality of life 2007	Average index of quality of life in each Locality in 2007	Survey of Quality of Life in Bogotá 2007
Log of Average monthly expenditure 2007	Average log of monthly expenditure per household in each Locality in 2007	Survey of Quality of Life in Bogotá 2007
Fixed Variables		
Average socioeconomic classification (estrato)	Average socioeconomic classification in each UPZ	Mapas Bogota
Nearby transmilenio stations	Number of TransMilenio stations located within the UPZ or whose 400-meter catchment area intersects the UPZ boundaries	Mapas Bogota
Access to arterial roads	Number of arterial roads located within the UPZ or whose 400-meter buffer zone intersects the UPZ boundaries	Mapas Bogota
Other Variables		
Annual population estimates for each UPZ	Number of people who live in a certain UPZ in a specific year	Datos abiertos Bogota, Secretaría Distrital de Planeación (District Planning Secretariat)

B Results without smoothed crime rates

While personal theft, motorcycle theft, and sexual violence estimates are similar with and without Empirical Bayes smoothing, two results are sensitive to this choice: homicide loses statistical significance in the unsmoothed specification, while vehicle theft becomes significant only without smoothing. Given that homicide is the crime category with the fewest observations per UPZ-quarter, and therefore the most susceptible to the small-population variance inflation that Empirical Bayes smoothing is designed to correct, the smoothed specification is preferred as the baseline. Nonetheless, this sensitivity should be interpreted as a limitation on the precision, rather than the direction, of the homicide effect, since the estimate remains positive across both specifications.

Table B.1: Effect of the Opening of OXXO Stores on Different Types of Crime Rates (Without Smoothing)

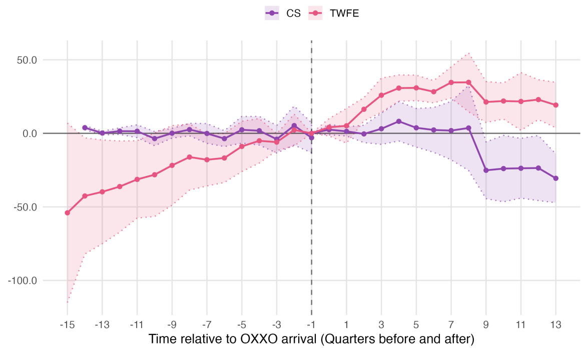
Crime rates	Personal Theft (1)	Sexual Violence (2)	Homicide (3)	Motorcycle Theft (4)	Vehicle Theft (5)
Has OXXO	2.215	-0.947^{*}	0.166	-0.265	-0.200
	(7.199)	(0.487)	(0.109)	(0.164)	(0.158)
Constant	31.73^{***}	2.736^{***}	0.544^{***}	1.647^{***}	1.504^{***}
	(0.265)	(0.056)	(0.017)	(0.025)	(0.024)
Observations	1,440	2,520	2,880	2,880	2,880
Treated (UPZ)	9	30	40	40	40
Never treated	81	60	50	50	50
R-squared	0.675	0.419	0.242	0.409	0.274
Mean never treated	55.531	2.501	0.292	1.819	1.176

Note: Standard errors in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.
TWFE considers fixed effects by unit and time, with standard errors clustered by UPZ. The dependent variable is the crime type rate per 10,000 inhabitants per UPZ. Moreover, the treatment variable is binary and indicates if the UPZ has at least one OXXO in its area. Personal theft rates are calculated only for the 2015–2018 period, and sexual violence rates only for 2015–2021, as I was unable to obtain SIEDCO data beyond 2018 and 2021, respectively. Always treated units are excluded from the analysis.

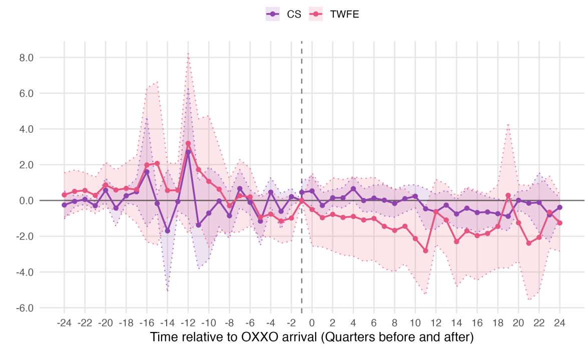
Table B.2: Comparison of ATT Estimates from TWFE and C&S Across Different Crime Types (Without Smoothing)

Crime rates	Personal Theft		Sexual Violence		Homicide		Motorcycle Theft		Vehicle Theft	
Method	TWFE	C&S	TWFE	C&S	TWFE	C&S	TWFE	C&S	TWFE	C&S
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
ATT	2.215	-2.009	-0.947^{*}	-0.196	0.166	0.082	-0.229	-0.121	-0.201^{**}	0.012
SE ATT	(7.199)	(6.831)	(0.487)	(0.414)	(0.109)	(0.092)	(0.166)	(0.259)	(0.093)	(0.149)
Observations	1,440	1,440	2,520	2,520	2,880	2,880	2,880	2,880	2,880	2,880

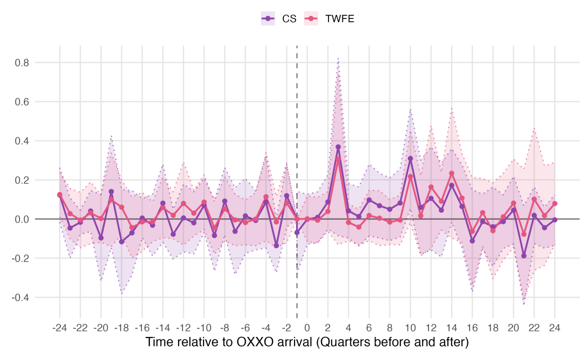
Note: Standard errors in parentheses. *** p<0.01, ** p<0.05, * p<0.1. ATT is the Average Treatment Effect on the Treated. For TWFE, this is the coefficient presented in Table B.1 and for C&S is the weighted average of the ATTs by cohort. Overall, the analysis covers 32 quarters for vehicle theft, motorcycle theft, and homicide, 28 quarters for sexual violence, and 16 quarters for personal theft, reflecting the availability of data for each crime category. Always treated units are excluded from the analysis.



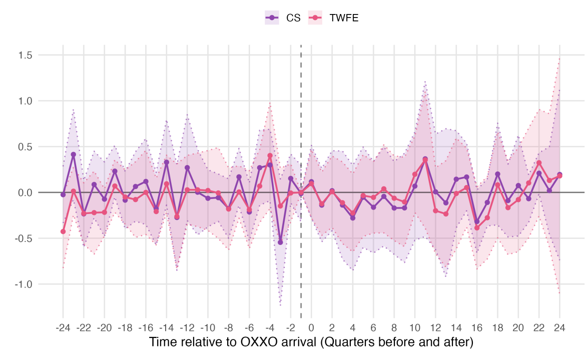
(a) Personal Theft



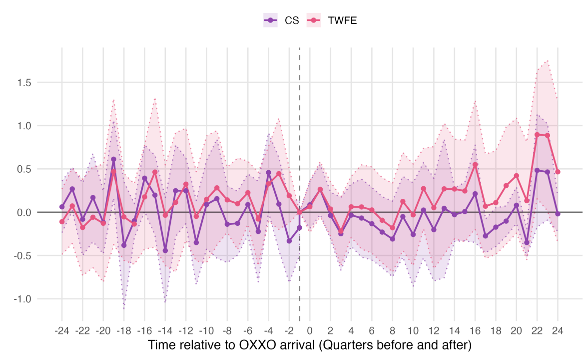
(b) Sexual Violence



(c) Homicide



(d) Motorcycle Theft



(e) Vehicle Thefts

Figure B.1: Event Study Analysis: TWFE vs C&S

Note: The rate of each crime is per 10,000 inhabitants. Rates are not smoothed.

Table B.3: Effect of the opening of OXXO stores on different types of crime rates in terms of treatment intensity (Without Smoothing)

Crime rates	Personal Theft (1)	Sexual Violence (2)	Homicide (3)	Motorcycle Theft (4)	Vehicle Theft (5)
Number of OXXO stores	−15.72	−0.580**	0.130	−0.252	−0.236
	(11.870)	(0.281)	(0.080)	(0.173)	(0.164)
Constant	48.19***	2.715***	0.509***	1.583***	1.490***
	(4.638)	(0.043)	(0.061)	(0.058)	(0.045)
Observations	1,440	2,520	2,880	2,880	2,880
Treated (UPZ)	9	30	40	40	40
Never treated	81	60	50	50	50
R-squared	0.715	0.419	0.292	0.411	0.290
Mean never treated	55.531	2.501	0.292	1.819	1.176

Note: Standard errors in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.
TWFE considers fixed effects by unit and time, with standard errors clustered by UPZ. The dependent variable is the crime type rate per 10,000 inhabitants per UPZ. Rates are not smoothed. Moreover, the treatment variable is continuous and indicates the number of OXXO stores in the UPZ. Personal theft rates are calculated only for the 2015–2018 period, and sexual violence rates only for 2015–2021, as I was unable to obtain SIEDCO data beyond 2018 and 2021, respectively. Always treated units are excluded from the analysis.

C Staggered controls of hard-discount store chains

Table C.1: Means differences of hard discount store staggered controls for the 2018-4 cohort

	Still without treatment	**Treated**	**Means Difference**
Has D1s	0.477 (0.054)	0.477 (0.079)	0.172***
Number of D1s	1.233 (0.214)	3.250 (0.543)	2.017***
Has Aras	0.256 (0.047)	0.714 (0.087)	0.458***
Number of Aras	0.384 (0.089)	1.429 (0.264)	0.216***
Has J&B	0.062 (0.256)	0.278 (1.202)	0.216***
Number of J&B	0.895 (0.050)	1.964 (0.096)	1.069
UPZs	86	28	114

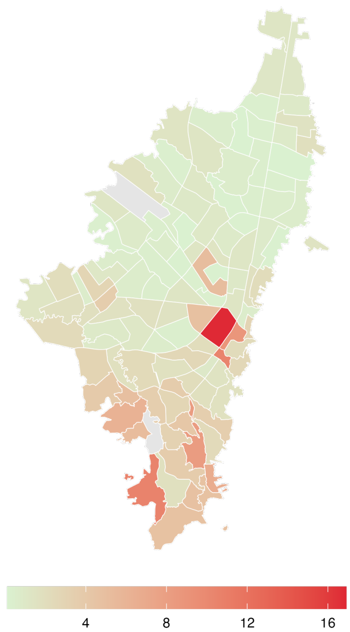
Note: Significance: *** p<0.01, ** p<0.05, * p<0.1.

Table C.2: Means differences of hard discount store staggered controls for the 2022-4 cohort

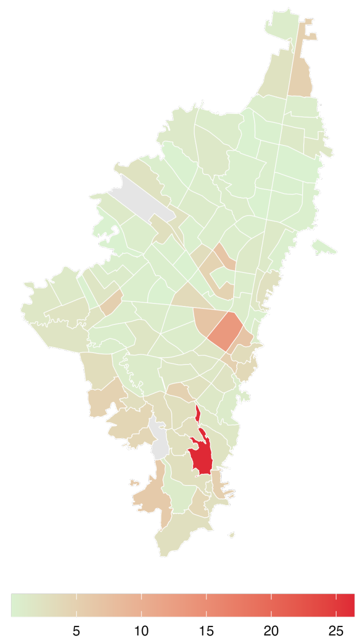
	Still without treatment	**Treated**	**Means Difference**
Has D1s	0.491 (0.067)	0.881 (0.042)	0.390***
Number of D1s	1.053 (0.191)	4.458 (0.512)	3.405***
Has Aras	0.298 (0.061)	0.627 (0.063)	0.329***
Number of Aras	0.386 (0.089)	1.712 (0.287)	1.326***
UPZs	57	59	116

Nota: Significance: *** p<0.01, ** p<0.05, * p<0.1. J&B does not appear because all stores closed in 2022.

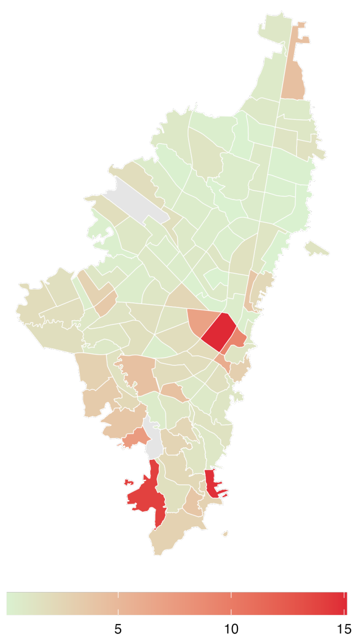
D Distribution of other crimes in Bogotá over time



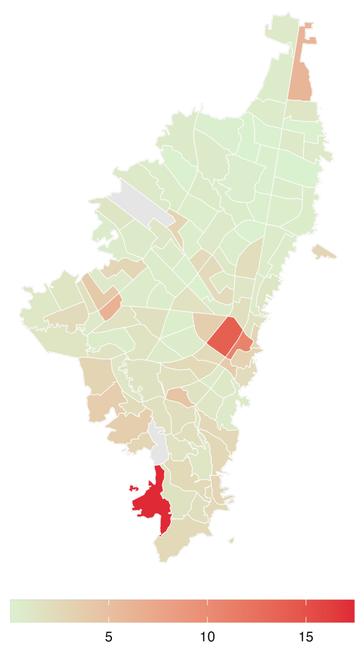
(a) 2015



(b) 2017

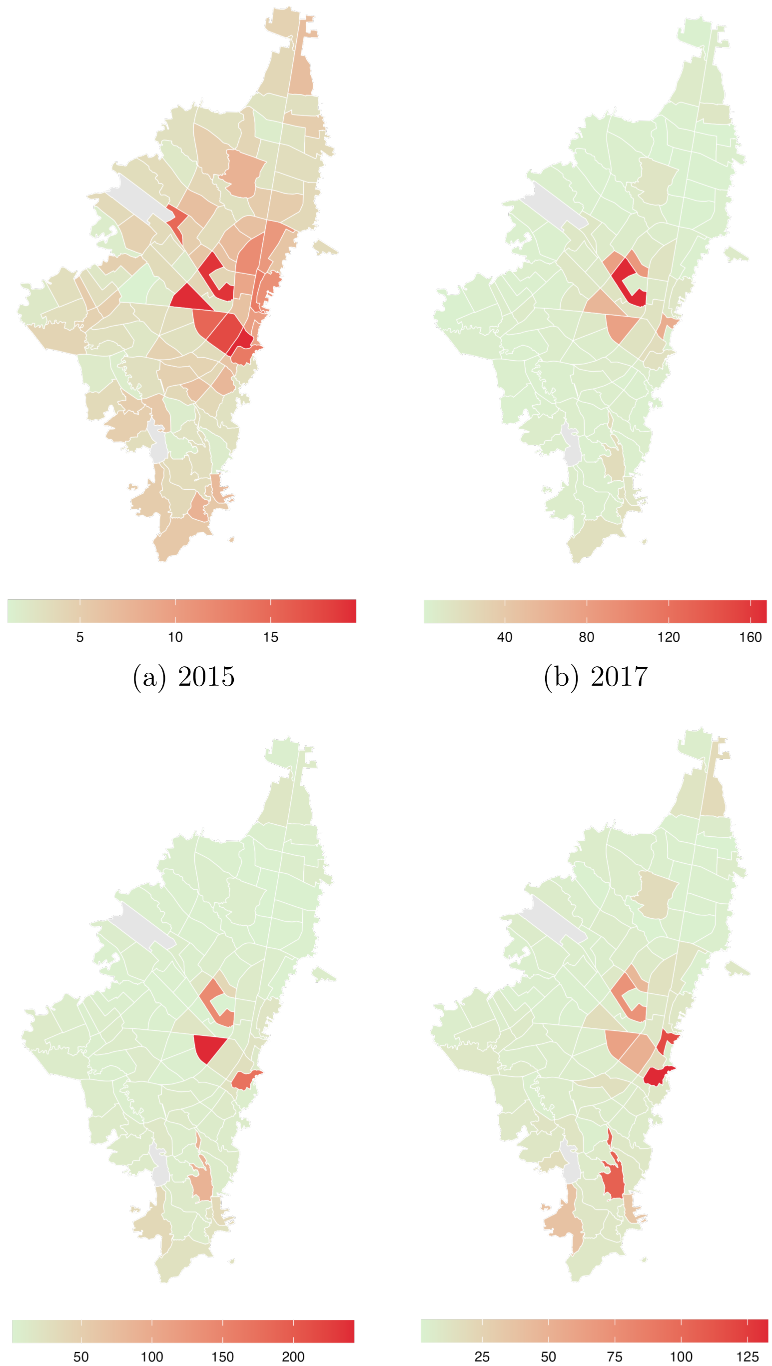


(c) 2019



(d) 2021

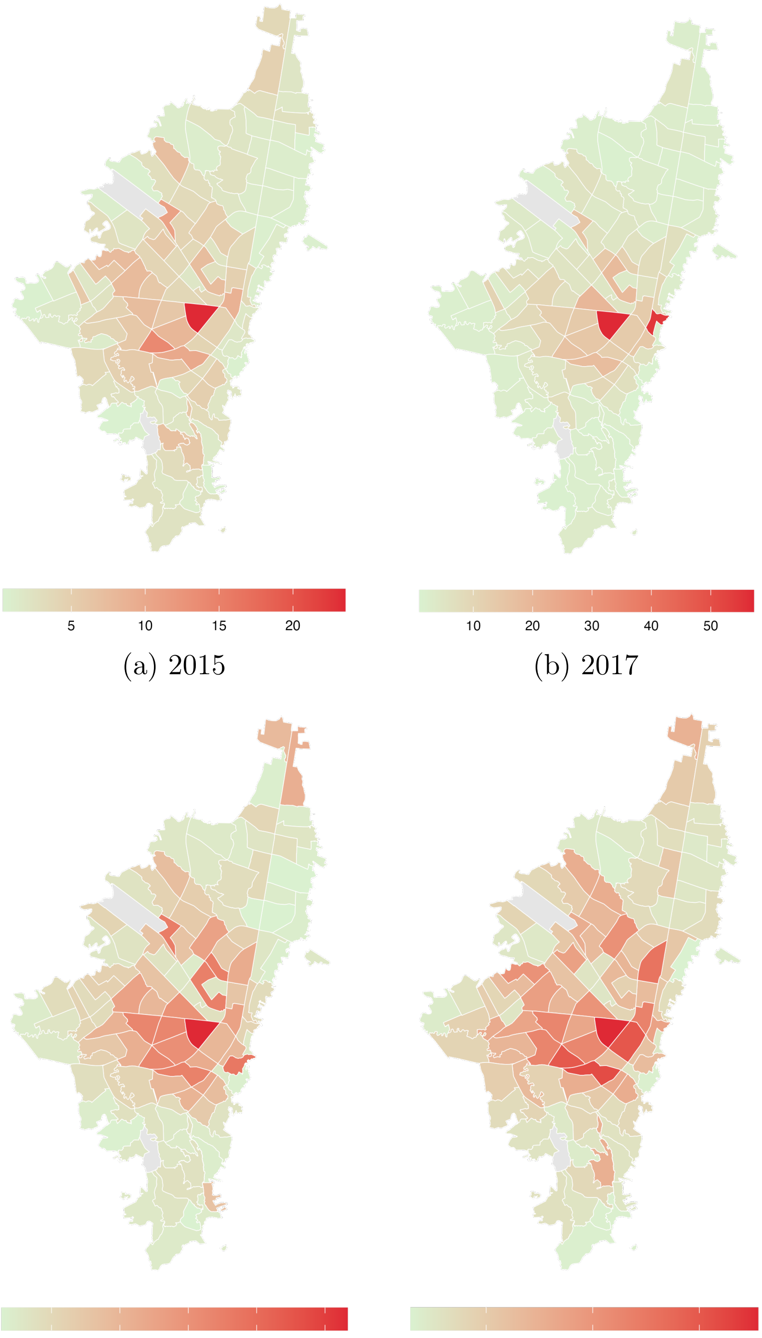
Figure D.1: Evolution of homicide rates by UPZ



(a) 2015 (b) 2017

(c) 2019 (d) 2021

Figure D.2: Evolution of sexual violence rates by UPZ



(a) 2015 (b) 2017

(c) 2019 (d) 2021

Figure D.3: Evolution of vehicle theft rates by UPZ

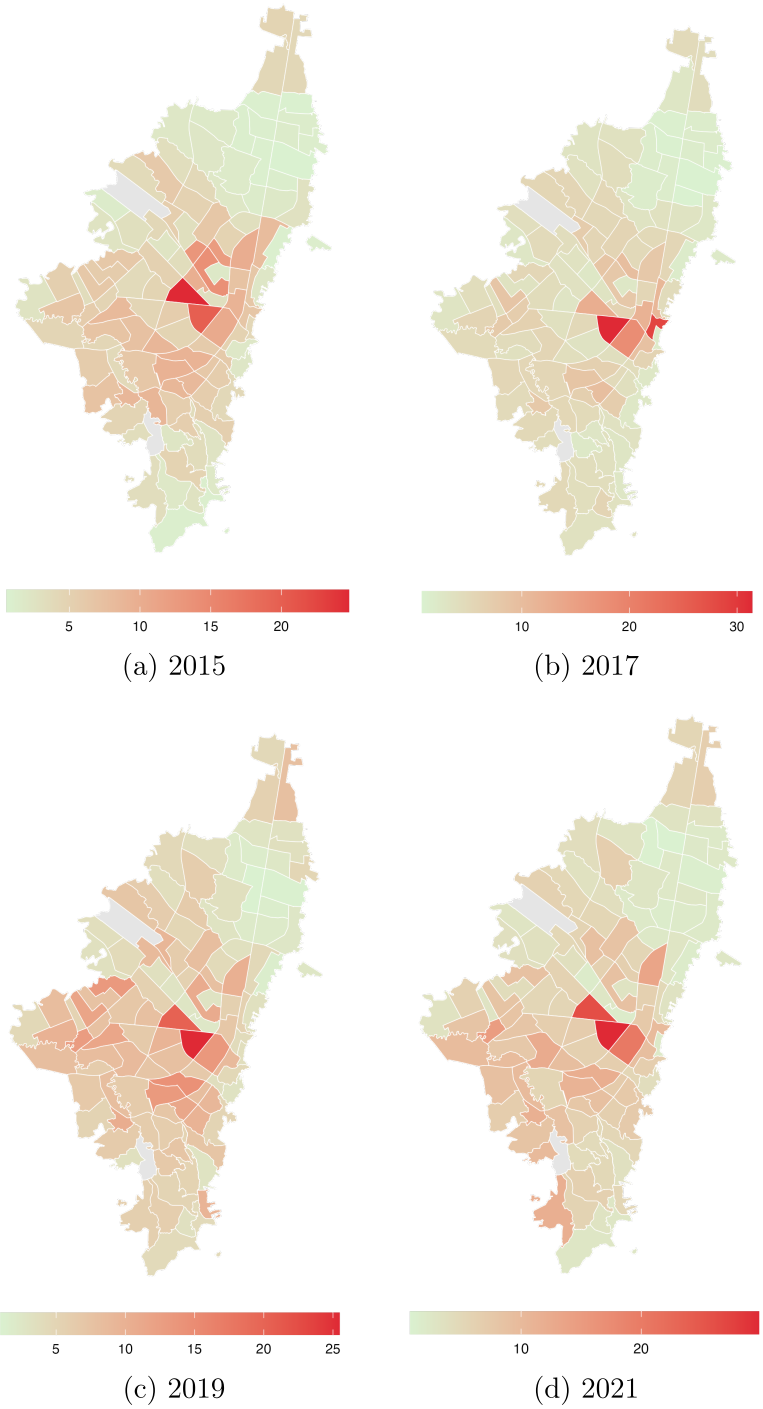


Figure D.4: Evolution of motorcycle theft rates by UPZ

E Results with hard-discount stores as controls

Table E.1: Effect of the opening of OXXO stores on different type of crime rates with staggered controls of hard-discount stores

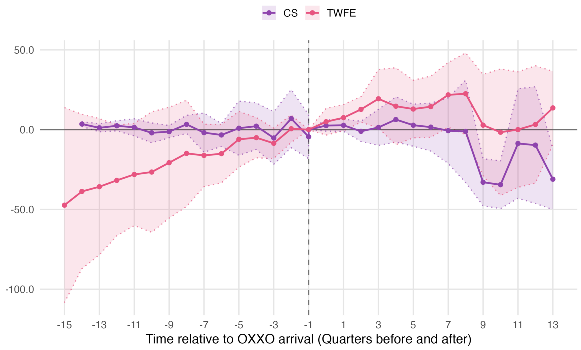
Crime rates	Personal Theft (1)	Sexual Violence (2)	Homicide (3)	Motorcycle Theft (4)	Vehicle Theft (5)
Has OXXO	2.095	-0.320	0.086^{***}	-0.112	-0.201^{**}
	(6.611)	(0.204)	(0.032)	(0.080)	(0.093)
Constant	33.65^{***}	2.405^{***}	0.454^{***}	1.493^{***}	1.309^{***}
	(2.800)	(0.097)	(0.013)	(0.023)	(0.029)
Chain Stores	yes	yes	yes	yes	yes
Observations	1,440	2,520	2,880	2,880	2,880
Treated (UPZ)	9	30	40	40	40
Never treated	81	60	50	50	50
R-squared	0.662	0.419	0.395	0.611	0.586
Mean never treated	55.531	2.167	0.495	1.643	1.153

Note: Standard errors in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.
TWFE considers fixed effects by unit and time, with standard errors clustered by UPZ. The dependent variable is the crime type rate per 10,000 inhabitants per UPZ. Moreover, the treatment variable is binary and indicates if the UPZ has at least one OXXO in its area. Personal theft rates are calculated only for the 2015–2018 period, and sexual violence rates only for 2015–2021, as I was unable to obtain SIEDCO data beyond 2018 and 2021, respectively. Always treated units are excluded from the analysis. Hard discount store controls include the presence of D1, Ara, and Justo & Bueno in the quarter before the analysis.

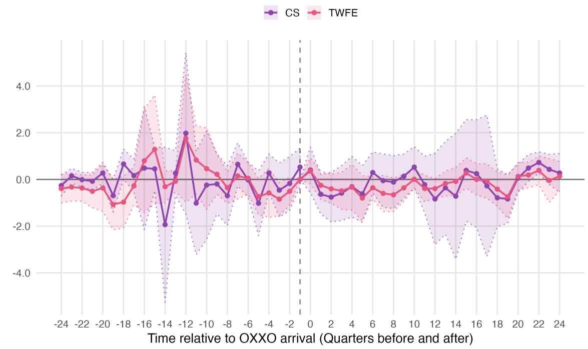
Table E.2: Comparison of ATT Estimates from TWFE and C&S Across Different Crime Types

Crime rates	Personal Theft		Sexual Violence		Homicide		Motorcycle Theft		Vehicle Theft	
Method	TWFE	C&S	TWFE	C&S	TWFE	C&S	TWFE	C&S	TWFE	C&S
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
ATT	2.095	−2.319	−0.320	0.059	0.086***	0.069	−0.112	−0.017	−0.201**	0.014
SE ATT	(6.611)	(6.445)	(0.204)	(0.365)	(0.032)	(0.085)	(0.080)	(0.165)	(0.093)	(0.221)
Observations	1,440	1,440	2,520	2,880	2,880	2,880	2,880	2,880	2,880	2,880

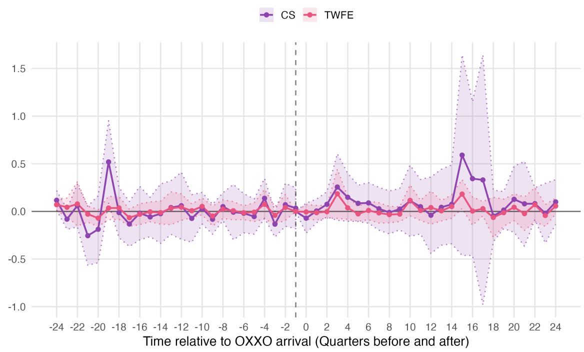
Note: Standard errors in parentheses. *** $p<0.01$, ** $p<0.05$, * $p<0.1$. ATT is the Average Treatment Effect on the Treated. For TWFE, this is the coefficient presented in Table E.1 and for C&S is the weighted average of the ATTs by cohort. In total, there are 32 quarters that represent the totality of the period of analysis. Always treated units are excluded from the analysis.



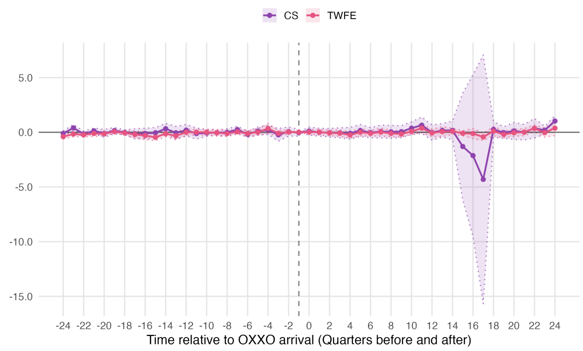
(a) Personal Theft



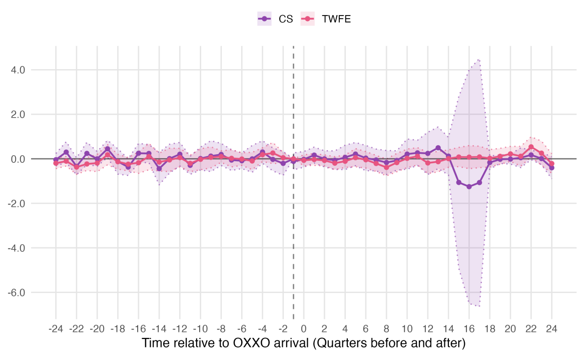
(b) Sexual Violence



(c) Homicide



(d) Motorcycle Theft



(e) Vehicle Thefts

Figure E.1: Event Study Analysis: TWFE Estimates

Note: The rate of each crime is per 10,000 inhabitants. Always treated UPZs are excluded from the analysis.

Table E.3: Effect of the opening of OXXO stores on different types of crime rates in terms of treatment intensity with staggered controls of hard-discount stores

Crime rates	Personal Theft (1)	Sexual Violence (2)	Homicide (3)	Motorcycle Theft (4)	Vehicle Theft (5)
Number of OXXO stores	2.095	-0.239^{*}	0.037^{***}	-0.025	-0.052
	(6.611)	(0.132)	(0.015)	(0.029)	(0.036)
Constant	33.65^{***}	2.404^{***}	0.458^{***}	1.486^{***}	1.297^{***}
	(2.800)	(0.095)	(0.012)	(0.021)	(0.030)
Chain Stores	yes	yes	yes	yes	yes
Observations	1,440	2,520	2,880	2,880	2,880
Treated (UPZ)	9	30	40	40	40
Never treated	81	60	50	50	50
R-squared	0.662	0.419	0.395	0.611	0.585
Mean never treated	55.531	2.167	0.495	1.643	1.153

Note: Standard errors in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.
TWFE considers fixed effects by unit and time, with standard errors clustered by UPZ. The dependent variable is the crime type rate per 10,000 inhabitants per UPZ. Moreover, the treatment variable is continuous and indicates the number of OXXO stores in the UPZ. Personal theft rates are calculated only for the 2015–2018 period, and sexual violence rates only for 2015–2021, as I was unable to obtain SIEDCO data beyond 2018 and 2021, respectively. Always treated units are excluded from the analysis. Hard discount store controls include the presence of D1, Ara, and Justo & Bueno in the quarter before the analysis. These covariates are also defined as continuous variables to maintain consistency with the treatment intensity specification.

F Results with always treated UPZs

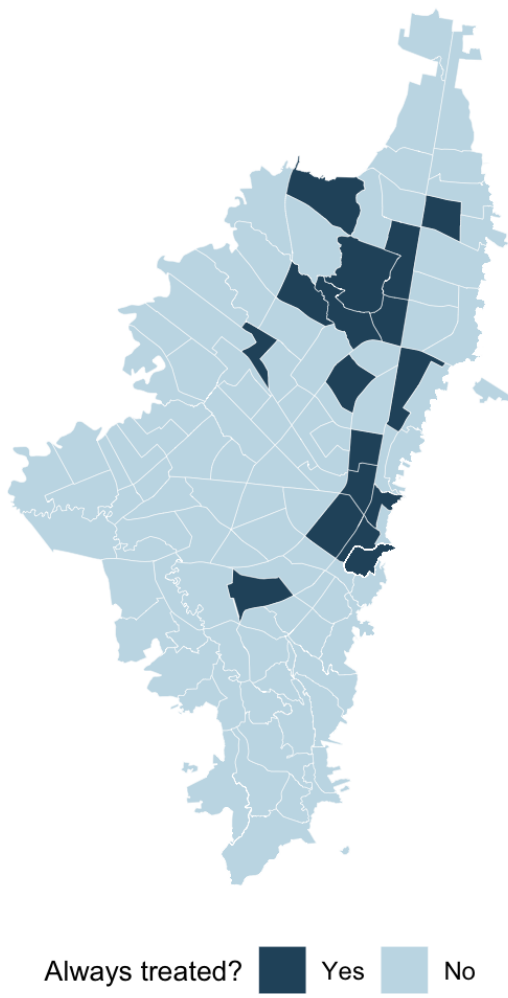


Figure F.1: Always Treated UPZs

Table F.1: Effect of OXXO Store Openings on Different Types of Crime Rates

Crime rates	Personal Theft (1)	Sexual Violence (2)	Homicide (3)	Motorcycle Theft (4)	Vehicle Theft (5)
Has OXXO	−7.336	−0.738**	0.070**	−0.067	−0.115
	(7.804)	(0.311)	(0.031)	(0.078)	(0.089)
Constant	42.23***	2.568***	0.461***	1.524***	1.588***
	(1.481)	(0.080)	(0.009)	(0.023)	(0.026)
Observations	1,712	2,996	3,424	3,424	3,424
Treated (UPZ)	26	30	57	57	57
Never treated	81	60	50	50	50
R-squared	0.696	0.433	0.486	0.410	0.552
Mean never treated	55.531	2.167	0.495	1.643	1.153

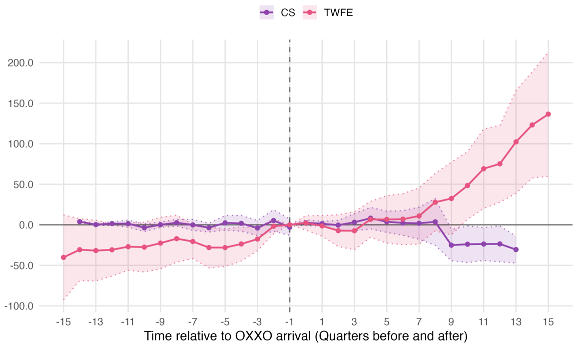
Note: Standard errors in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.
TWFE considers fixed effects by unit and time, with standard errors clustered by UPZ. The dependent variable is the smoothed crime type rate per 10,000 inhabitants per UPZ. Moreover, the treatment variable is binary and indicates if the UPZ has at least one OXXO in its area. Personal theft rates are calculated only for the 2015–2018 period, and sexual violence rates only for 2015–2021, as I was unable to obtain SIEDCO data beyond 2018 and 2021, respectively.

Figure 8 compares how the C&S and TWFE ES behave in each crime type. These graphs reveal that both have a similar evolution in most crimes, which again indicate that there is no bias in the TWFE specification. Additionally, C&S ES present more robust results as all crimes satisfy the parallel trends assumption, this indicates that Sexual Violence and personal theft C&S estimates are more reliable.

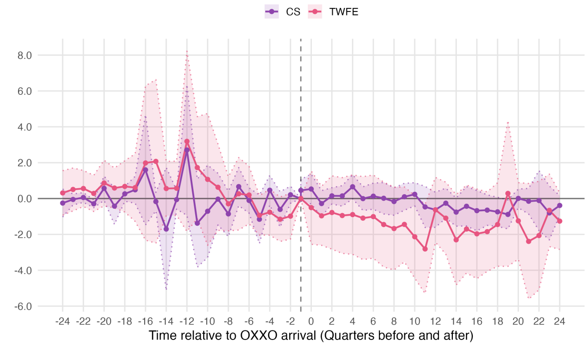
Table F.2: Comparison of ATT Estimates from TWFE and C&S Across Different Crime Types with Always Treated Observations

Crime rates	Personal Theft		Sexual Violence		Homicide		Motorcycle Theft		Vehicle Theft	
Method	TWFE	C&S	TWFE	C&S	TWFE	C&S	TWFE	C&S	TWFE	C&S
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
ATT	-7.435	-1.917	-0.738^{**}	0.008	0.070^{**}	0.080	-0.067	-0.029	-0.115	-0.058
SE ATT	(7.837)	(6.699)	(0.311)	(0.224)	(0.078)	(0.058)	(0.078)	(0.165)	(0.089)	(0.119)
Observations	1,712	1,440	2,996	2,520	3,424	2,880	3,424	2,880	3,424	2,880

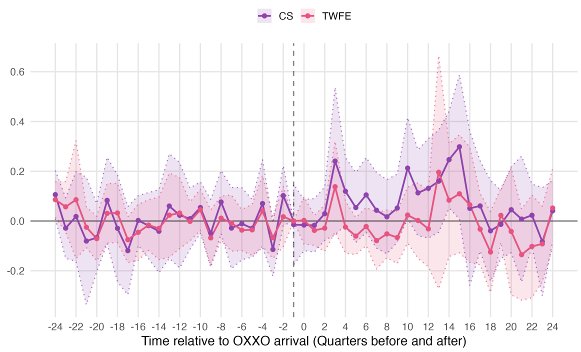
Note: Standard errors in parentheses. *** p<0.01, ** p<0.05, * p<0.1. ATT is the Average Treatment Effect on the Treated. For TWFE, this is the coefficient presented in Table F.1 and for C&S is the weighted average of the ATTs by cohort. Overall, the analysis covers 32 quarters for vehicle theft, motorcycle theft, and homicide, 28 quarters for sexual violence, and 16 quarters for personal theft, reflecting the availability of data for each crime category.



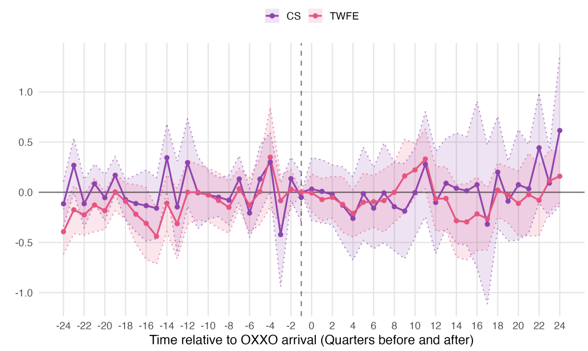
(a) Personal Theft



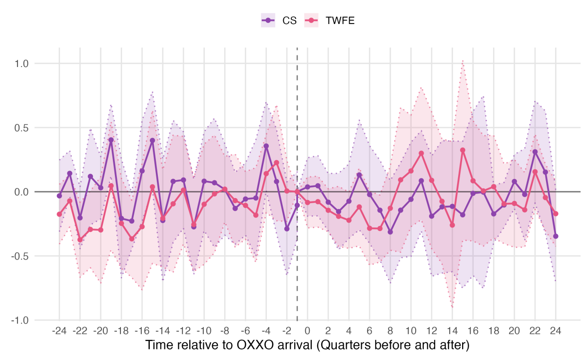
(b) Sexual Violence



(c) Homicide



(d) Motorcycle Theft



(e) Vehicle Thefts

Figure F.2: Event Study Analysis: TWFE vs. Callaway & Sant'Anna (C&S) Estimates
Note: The rate of each crime is per 10,000 inhabitants.

Personal theft is the only crime category that exhibits a clear temporal trend in both the TWFE and C&S event studies after the opening of the first OXXO store; however, the estimated trends differ in direction across the two specifications. These differences can be explained by the nature of each event study approach. The TWFE specification only considers treated units and includes always-treated observations. In fact, during this period (from 2015 to 2018), 65% of the 26 treated UPZs were already treated at the beginning of the sample, leaving only nine UPZs that received treatment after 2015-Q1. In contrast, the C&S excludes always-treated units and compares only these nine newly treated UPZs with not-yet-treated ones. This indicates that the positive trend observed in the TWFE specification is largely driven by the always-treated UPZs, whereas the control units in the C&S one offset these differences by providing a more appropriate counterfactual. Overall, this evidence reinforces the interpretation that the TWFE coefficient for personal theft is biased and does not accurately identify the causal effect of OXXO entry.

Finally, Table F.3 presents the TWFE estimates using treatment intensity. The results differ substantially from those reported in Table F.1. In particular, personal theft becomes positive and statistically significant, a finding that is consistent with the hypothesis that OXXO stores may act as crime attractors. However, the estimated magnitude is considerably large and raises concerns about the robustness of the result, particularly given the substantial variation in the estimated effect for personal theft across specifications. Motorcycle theft also becomes statistically significant while retaining a negative coefficient.

Table F.3: Effects of OXXO Store Openings on Different Crime Rates under the Treatment-Intensity Specification with Always Treated Observations

Crime rates	personal theft (1)	Sexual Violence (2)	Homicide (3)	Motorcycle Theft (4)	Vehicle Theft (5)
Number of OXXO stores	50.33^{**}	-0.640^{**}	0.047	-0.076^{*}	0.040
	(21.58)	(0.293)	(0.035)	(0.0432)	(0.033)
Constant	25.430^{***}	2.881^{***}	0.547^{***}	1.642^{***}	1.510^{***}
	(6.607)	(0.202)	(0.020)	(0.024)	(0.018)
Observations	1,712	2,996	3,424	3,424	3,424
Treated (UPZ)	26	47	57	57	57
Never treated	81	60	50	50	50
R-squared	0.714	0.393	0.292	0.41	0.288
Mean never treated	55.531	2.167	0.495	1.643	1.153

Note: Standard errors in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.
TWFE considers fixed effects by unit and time, with standard errors clustered by UPZ. The dependent variable is the crime type rate per 10,000 inhabitants per UPZ. Moreover, the treatment variable is continuous and indicates the number of OXXO stores in the UPZ. Personal theft rates are calculated only for the 2015–2018 period, and sexual violence rates only for 2015–2021, as I was unable to obtain SIEDCO data beyond 2018 and 2021, respectively.